

**ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS**

Inferential Statistics for Business

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

2027 EDITION

LET ME INTRODUCE MYSELF.

About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- ✓ My name is **Jovan Samardzic**, I am 26 years old
- ✓ Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- ✓ Recipient of the **Science@Leuven Scholarship**

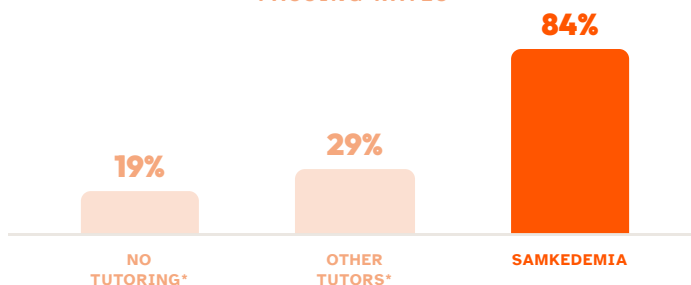
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 +32 490 10 08 21

 samkedia

 jovan@samkedia.com

 samkedia.com

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JOVAN SAMARDZIC PR BEOGRAD (STARI GRAD)

PIB: 115342864 MB: 68292042

 +32490100821

 samkedemia

 jovan@samkedemia.com

 samkedemia.com

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Chapter 1 - Introduction

? Question 1

A portfolio consists of a fraction p of investments in real estate and a fraction $1 - p$ of investments in stocks. The return on the investments in real estate is measured by X and the return on the investments in stocks is measured by Y . The return of the portfolio is measured by $R = pX + (1 - p)Y$.

Suppose that $\mu_X = 5\%$, $\sigma_X = 2.8\%$, $\mu_Y = 11.7\%$, $\sigma_Y = 8.6\%$, and $\rho = -0.16$.

- (a) Give an economic explanation for the negative correlation.
- (b) Calculate the expectation and standard deviation of R in the case $p = 0.7$.
- (c) Determine the value of p at which $\text{Var}[R]$ attains its minimum. (In mathematical finance, $\text{Var}[R]$ is a measure for the risk of the portfolio.)

✓ Solution 1: (a) Provide a brief economic explanation for the negative correlation.

Real estate and stocks compete for the same investment money. When stock returns rise, investors tend to shift money out of real estate and into stocks, which can push real estate returns down - and the other way around when stocks do poorly.

Answer: This substitution between the two asset classes is what produces the negative correlation.

💡 Tip: Mean and Standard Deviation of a Linear Combination

For any linear combination $W = aX + bY + c$ of two random variables X and Y , the mean and standard deviation of W are:

$$\mu_W = a\mu_X + b\mu_Y + c, \quad \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho_{X,Y}\sigma_X\sigma_Y}$$

This holds generally - whether or not X and Y are normal, correlated, or independent. The constant c never affects the spread, only the mean. If X and Y are independent, $\rho_{X,Y} = 0$ and the cross term drops out.

✅ Solution 1: (b) Calculate the expectation and standard deviation of the portfolio return R for $p = 0.7$

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random real estate investment and a random stock investment, and record their returns.

- **Mother variables:**

X return on a random real estate investment (in %)

Y return on a random stock investment (in %)

- **Other random variables:**

$R = pX + (1 - p)Y$ return on the combined random portfolio

• **Given:**

$\mu_X = 0.05$	population mean return of all real estate investments
$\sigma_X = 0.028$	population standard deviation of return of all real estate investments
$\mu_Y = 0.117$	population mean return of all stock investments
$\sigma_Y = 0.086$	population standard deviation of return of all stock investments
$\rho_{X,Y} = -0.16$	population correlation between all real estate and stock returns

.....

Since the problem does not state that X and Y are normal, we can find R 's mean and standard deviation directly, but cannot conclude a specific distribution for R .

Step 1: Find the mean of R .

With $a = p = 0.7$ and $b = 1 - p = 0.3$:

$$\begin{aligned}\mu_R &= a\mu_X + b\mu_Y = (0.7 \cdot 0.05) + (0.3 \cdot 0.117) \\ &= 0.035 + 0.0351 = 0.0701\end{aligned}$$

Step 2: Find the standard deviation of R .

First we find the variance:

$$\begin{aligned}\sigma_R^2 &= a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho_{X,Y}\sigma_X\sigma_Y \\ &= (0.7)^2 \cdot (0.028)^2 + (0.3)^2 \cdot (0.086)^2 + 2 \cdot 0.7 \cdot 0.3 \cdot (-0.16) \cdot 0.028 \cdot 0.086 \\ &= 0.0008879824\end{aligned}$$

Then take the square root:

$$\sigma_R = \sqrt{0.0008879824} \approx 0.0298$$

Answer: The expected return is $\mu_R = 0.0701$ (7.01%), and the standard deviation is $\sigma_R \approx 0.0298$ (2.98%).

💡 Tip: Minimizing a Quadratic Function

A quadratic function $g(p) = Ap^2 + Bp + C$ with $A > 0$ opens upward, so its minimum is at its vertex:

$$p^* = -\frac{B}{2A}$$

Equivalently, this is exactly where the first derivative $g'(p) = 2Ap + B$ equals zero - both methods always agree.

✅ Solution 1: (c) Find the fraction p that minimizes the portfolio risk (variance)

Step 1: Rewrite $\text{Var}(R)$ as a quadratic function of p .

With $a = p$ and $b = 1 - p$:

$$\text{Var}(R) = p^2\sigma_X^2 + (1-p)^2\sigma_Y^2 + 2p(1-p)\rho_{X,Y}\sigma_X\sigma_Y$$

Substituting the given values and expanding in p :

$$\text{Var}(R) = 0.00895056 p^2 - 0.01556256 p + 0.007396$$

We can minimize this quadratic in two equivalent ways:

Method 1: Vertex of a parabola.

Here $A = 0.00895056$ and $B = -0.01556256$:

$$p = -\frac{B}{2A} = -\frac{-0.01556256}{2 \cdot 0.00895056} = \frac{0.01556256}{0.01790112} \approx 0.8694$$

Method 2: Using derivatives.

Take the derivative of $\text{Var}(R)$ with respect to p and set it to zero:

$$\frac{d}{dp}\text{Var}(R) = 0.01790112p - 0.01556256 \Rightarrow p = \frac{0.01556256}{0.01790112} \approx 0.8694$$

The second derivative, 0.01790112, is positive, confirming this is a minimum.

Step 3: Find the mean and standard deviation of R at this minimum-variance p .

With $a = p \approx 0.86936$ and $b = 1 - p \approx 0.13064$:

$$\mu_R = a\mu_X + b\mu_Y = (0.86936 \cdot 0.05) + (0.13064 \cdot 0.117) \approx 0.05875$$

$$\sigma_R^2 = 0.00895056p^2 - 0.01556256p + 0.007396 \approx 0.0006311$$

$$\sigma_R \approx \sqrt{0.0006311} \approx 0.02512$$

Answer: The portfolio risk is minimized at $p \approx 0.8694$, i.e. about 86.94% invested in real estate. At this p , $\mu_R \approx 0.05875$ (5.875%) and $\sigma_R \approx 0.02512$ (2.512%).

Question 2

Let X measure the taxable income of the wife in a randomly selected heterosexual married couple and Y measure the taxable income of the husband in the same married couple. Assume that (X, Y) has a bivariate normal distribution with

$$\mu_X = \text{€}1734, \quad \sigma_X = \text{€}366, \quad \mu_Y = \text{€}1921, \quad \sigma_Y = \text{€}453, \quad \rho_{X,Y} = 0.45.$$

- What is the probability distribution of the total taxable income of a randomly chosen married couple?
- Women must pay a tax of 30% and men a tax of 40%. What is the probability that the net income of the wife is larger than the net income of the husband in a randomly selected married couple?

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ **Solution 2: (a) What is the probability distribution of the total taxable income of a randomly chosen married couple?**

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random heterosexual married couple and record the taxable incomes of the wife and husband.

- **Mother variables:**

X taxable income of the wife in a random married couple (in €)

Y taxable income of the husband in a random married couple (in €)

- **Other random variables:**

$T = X + Y$ total taxable income of a random married couple

- **Given:** (X, Y) is bivariate normal, with population parameters:

$\mu_X = 1734$ population mean taxable income of all wives

$\sigma_X = 366$ population standard deviation of taxable income of all wives

$\mu_Y = 1921$ population mean taxable income of all husbands

$\sigma_Y = 453$ population standard deviation of taxable income of all husbands

$\rho_{X,Y} = 0.45$ population correlation between all wives' and husbands' incomes

.....
Since T is a linear combination with $a = 1$, $b = 1$, $c = 0$, it is also exactly normal - we just need its mean and standard deviation:

$$\mu_T = 1734 + 1921 = 3655$$

$$\sigma_T = \sqrt{366^2 + 453^2 + 2 \cdot 366 \cdot 453 \cdot 0.45} \approx 698.84$$

Answer: The total taxable income follows $T \sim \mathcal{N}(3655, 698.84)$ (in €).

💡 Tip: Comparing Two Variables

To find a probability comparing two variables (e.g. $P(X > Y)$), move everything to one side first, turning it into a statement about their difference:

$$P(X > Y) = P(X - Y > 0)$$

Then find the distribution of $X - Y$ using the linear combination formula for two normal variables.

✅ Solution 2: (b) What is the probability that the net income of the wife is larger than the net income of the husband?

Step 1: Rewrite the comparison as a statement about the difference.

The wife pays 30% tax, so her net income is $0.7X$; the husband pays 40% tax, so his net income is $0.6Y$. We move everything to one side:

$$P(0.7X > 0.6Y) = P(0.7X - 0.6Y > 0)$$

Define $D = 0.7X - 0.6Y$, the difference in net incomes.

Step 2: Find the distribution of the difference.

This is a linear combination with $a = 0.7$, $b = -0.6$, $c = 0$:

$$\mu_D = 0.7 \cdot 1734 - 0.6 \cdot 1921 = 61.2$$

$$\sigma_D = \sqrt{(0.7)^2 \cdot 366^2 + (-0.6)^2 \cdot 453^2 + 2 \cdot 0.7 \cdot (-0.6) \cdot 366 \cdot 453 \cdot 0.45} \approx 277.20$$

So the difference is also normally distributed:

$$D \sim \mathcal{N}(61.2, 277.20)$$

Step 3: Compute the probability.

Since μ_D is positive, the wife's net income is higher than the husband's on average, so we expect

$P(D > 0)$ above 0.5:

$$P(D > 0) = \text{normalcdf}(0, \text{E}99, 61.2, 277.20) \approx 0.5874$$

Answer: The probability that the wife's net income is larger than the husband's is 0.5874.

Question 3

The weights of butter dishes that leave some factory are normally distributed with mean 250 g and standard deviation 4 g. Let X and Y measure the weights of two independently and randomly chosen butter dishes from the production.

- (a) What is the value of $P(X > Y)$?
- (b) Calculate $P(|X - Y| > 4)$.
- (c) Determine $P(\max(X, Y) < 250)$.

Tip: Comparing Two Variables

To find a probability comparing two variables (e.g. $P(X > Y)$), move everything to one side first, turning it into a statement about their difference:

$$P(X > Y) = P(X - Y > 0)$$

Then find the distribution of $X - Y$ using the linear combination formula for two normal variables.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ Solution 3: (a) What is the value of $P(X > Y)$?

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random butter dish from the factory and record its weight.

- **Mother variables:**

X weight of a random butter dish (in g)

Y weight of a second, independent random butter dish (in g)

- **Other random variables:**

$D = X - Y$ difference in weight between the two random butter dishes (in g)

- **Given:**

$\mu_X = \mu_Y = 250$ population mean weight of all butter dishes

$\sigma_X = \sigma_Y = 4$ population standard deviation of weight of all butter dishes

X, Y are independent

.....

Step 1: Translate the comparison into a statement about the difference.

We move everything to one side:

$$P(X > Y) = P(X - Y > 0)$$

Define $D = X - Y$. Since X and Y are independent, this is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho_{X,Y} = 0$.

Step 2: Find the distribution of D , and compute the probability.

Plug in the given means and standard deviations:

$$\mu_D = 250 - 250 = 0$$

$$\sigma_D = \sqrt{4^2 + 4^2} = \sqrt{32} \approx 5.657$$

So $D \sim \mathcal{N}(0, 5.657)$. Since this distribution is symmetric around 0, the probability of a value above 0 is exactly half:

$$P(X > Y) = P(D > 0) = \text{normalcdf}(0, \text{E99}, 0, 5.657) = 0.5$$

Answer: $P(X > Y) = 0.5$.

✓ Solution 3: (b) Calculate $P(|X - Y| > 4)$

Step 1: Translate the event into two one-sided pieces.

"Differs by more than 4 g" doesn't say which dish is heavier - either X could exceed Y by more than 4 g, or Y could exceed X by more than 4 g. These are two separate, non-overlapping events, so we add their probabilities instead of using absolute value notation:

$$P(D > 4) + P(D < -4)$$

where $D = X - Y \sim \mathcal{N}(0, 5.657)$ from part (a).

Step 2: Compute the probability.

By symmetry, both tails carry the same probability:

$$P(D > 4) = \text{normalcdf}(4, \text{E99}, 0, 5.657) \approx 0.23975$$

$$P(D < -4) = \text{normalcdf}(-\text{E99}, -4, 0, 5.657) \approx 0.23975$$

Adding them:

$$P(D > 4) + P(D < -4) \approx 0.23975 + 0.23975 = 0.4795$$

Answer: $P(|X - Y| > 4) \approx 0.4795$.

 Tip: Probability of a Maximum

The statement "the maximum of two values is less than a number" is a concise way of saying that **both** of the individual values must be less than that number.

$$P(\max(X, Y) < a) = P(X < a \text{ and } Y < a)$$

Furthermore, if X and Y are independent, this simplifies to the product of their individual probabilities:

$$P(\max(X, Y) < a) = P(X < a) \cdot P(Y < a)$$

 Solution 3: (c) Determine $P(\max(X, Y) < 250)$ **Step 1: Rewrite the maximum condition as a product.**

Since the choices of butter dishes are independent, we can multiply their probabilities:

$$P(\max(X, Y) < 250) = P(X < 250 \text{ and } Y < 250) = P(X < 250) \cdot P(Y < 250)$$

Step 2: Compute the probability.

First, we find the probability for a single dish, $P(X < 250)$. Since the distribution $X \sim N(250, 4)$ is symmetric around its mean of 250, the probability of being less than the mean is exactly 0.5:

$$P(X < 250) = 0.5$$

Now we can calculate the final probability:

$$P(\max(X, Y) < 250) = 0.5 \cdot 0.5 = 0.25$$

Answer: The probability that the maximum weight of the two dishes is less than 250 g is 0.25.

Question 4

The random pair (W, L) measures the (width, length) of a random rectangular piece of building site and is bivariate normally distributed with $\mu_W = 30$ m, $\sigma_W = 2$ m, $\mu_L = 50$ m, $\sigma_L = 3$ m, and correlation $\rho_{W,L}$. The calculated perimeter is $X = 2(W + L)$.

- Calculate $P(X \leq 165)$ in case $\rho_{W,L} = 0$.
- Calculate $P(X \leq 165)$ in case $\rho_{W,L} = +0.5$.
- Calculate $P(X \leq 165)$ in case $\rho_{W,L} = -0.5$.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ Solution 4: (a) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = 0$

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random rectangular building site and record its width and length.
- **Mother variables:**

W	width of a random building site (in m)
L	length of the same random building site (in m)

- **Other random variables:**

$$X = 2W + 2L \quad \text{perimeter of the random building site (in m)}$$

- **Given:** (W, L) is bivariate normal, with population parameters:

$$\mu_W = 30 \quad \text{population mean width of all building sites}$$

$$\sigma_W = 2 \quad \text{population standard deviation of width of all building sites}$$

$$\mu_L = 50 \quad \text{population mean length of all building sites}$$

$$\sigma_L = 3 \quad \text{population standard deviation of length of all building sites}$$

$$\rho_{W,L} \quad \text{population correlation between width and length of all building sites}$$

.....

Bivariate normal just means W and L are each individually normal, linked by $\rho_{W,L}$. Because of this, any linear combination of them - including the perimeter $X = 2W + 2L$ - is itself exactly normal too, no matter how strongly W and L are correlated. This is what makes all three parts solvable with `normalcdf`, once we find X 's mean and standard deviation.

Step 1: Identify a , b , and c .

The perimeter is $X = 2W + 2L$, so $a = 2$, $b = 2$, $c = 0$.

Step 2: Find the mean.

The mean never depends on the correlation, so this value is the same for every part of this exercise. First the general formula, then plug in:

$$\mu_X = a\mu_W + b\mu_L + c = 2\mu_W + 2\mu_L = 2 \cdot 30 + 2 \cdot 50 = 160$$

Step 3: Find the standard deviation, as a general function of $\rho_{W,L}$.

Unlike the mean, the standard deviation does depend on the correlation - so instead of plugging

in a number right away, we keep $\rho_{W,L}$ symbolic, to reuse this same formula in parts (b) and (c):

$$\begin{aligned}\sigma_X(\rho_{W,L}) &= \sqrt{a^2\sigma_W^2 + b^2\sigma_L^2 + 2ab\sigma_W\sigma_L\rho_{W,L}} \\ &= \sqrt{2^2 \cdot 2^2 + 2^2 \cdot 3^2 + 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \rho_{W,L}} \\ &= \sqrt{16 + 36 + 48 \rho_{W,L}} = \sqrt{52 + 48 \rho_{W,L}}\end{aligned}$$

So:

$$X \sim \mathcal{N}(160, \sqrt{52 + 48 \rho_{W,L}})$$

Step 4: Plug in $\rho_{W,L} = 0$ and compute the probability.

Substitute $\rho_{W,L} = 0$ into the formula from Step 3:

$$\sigma_X(0) = \sqrt{52 + 48 \cdot 0} = \sqrt{52} \Rightarrow X \sim \mathcal{N}(160, \sqrt{52})$$

Now we find the probability:

$$P(X \leq 165) = \text{normalcdf}(-E99, 165, 160, \sqrt{52}) \approx 0.7560$$

Answer: $P(X \leq 165) \approx 0.7560$.

✓ Solution 4: (b) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = 0.5$

Step 1: Plug in $\rho_{W,L} = 0.5$.

We reuse $X \sim \mathcal{N}(160, \sqrt{52 + 48 \rho_{W,L}})$ from part (a):

$$\sigma_X(0.5) = \sqrt{52 + 48 \cdot 0.5} = \sqrt{76} \Rightarrow X \sim \mathcal{N}(160, \sqrt{76})$$

Step 2: Compute the probability.

With X 's distribution in hand:

$$P(X \leq 165) = \text{normalcdf}(-E99, 165, 160, \sqrt{76}) \approx 0.7169$$

Answer: $P(X \leq 165) \approx 0.7169$.

✓ **Solution 4: (c) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = -0.5$**

Step 1: Plug in $\rho_{W,L} = -0.5$.

We reuse $X \sim \mathcal{N}(160, \sqrt{52 + 48 \rho_{W,L}})$ from part (a):

$$\sigma_X(-0.5) = \sqrt{52 + 48 \cdot (-0.5)} = \sqrt{28} \Rightarrow X \sim \mathcal{N}(160, \sqrt{28})$$

Step 2: Compute the probability.

With X 's distribution in hand:

$$P(X \leq 165) = \text{normalcdf}(-E99, 165, 160, \sqrt{28}) \approx 0.8276$$

Answer: $P(X \leq 165) \approx 0.8276$.

Question 5

In an elevator, there is a warning: "maximum 500 kg or 6 persons." Suppose that the body weights are normally distributed with mean 75 kg and standard deviation 15 kg. The body weight of a random person can be considered as independent of the body weights of other randomly chosen people.

What is the probability that 6 randomly chosen people weigh more than 500 kg?

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Solution 5:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random person and record their body weight.
- **Mother variable:**

X weight of a random person (in kg)

- **Other random variables:**

$W_6 = X_1 + \dots + X_6$ total weight of 6 random people (in kg)

• **Given:**

$\mu = 75$	population mean weight of all people
$\sigma = 15$	population standard deviation of weight of all people
$n = 6$	sample size

$$X \sim \mathcal{N}(\mu = 75, \sigma = 15)$$

$X_1, \dots, X_6$ are independent

.....

Step 1: Find the distribution of the total weight.

The weight of one person is normal. We're looking for the sum of 6 people's weights, so:

$$\left. \begin{array}{l} X_1, \dots, X_6 \sim \mathcal{N}(75, 15) \\ \text{independent} \end{array} \right\} \Rightarrow W_6 \sim \mathcal{N}(6 \cdot 75, \sqrt{6} \cdot 15) = \mathcal{N}(450, \sqrt{1350})$$

Step 2: Compute the probability.

Now that we know the distribution of W_6 , we can find the probability that it exceeds 500 kg:

$$P(W_6 > 500) = \text{normalcdf}(500, \text{E99}, 450, \sqrt{1350}) \approx 0.0868$$

Answer: The probability that 6 randomly chosen people weigh more than 500 kg is 0.0868.

Question 6

In a barbershop, the durations of the haircuts of men are normally distributed with mean 21 minutes and standard deviation 6 minutes, and the durations of the haircuts of women are normally distributed with mean 48 minutes and standard deviation 12 minutes. The durations of different haircuts can be assumed to be independent.

- There are 3 male customers in this barbershop. What is the probability that the total duration of their haircuts exceeds 1.5 hours?
- What is the probability that the average duration of the haircuts of 3 men is larger than 30 minutes?
- What is the probability that the total duration of the haircuts of 5 men is smaller than the total duration of the haircuts of 2 women?

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

✓ Solution 6: (a) Probability the total duration of 3 men's haircuts exceeds 1.5 hours

Before computing anything, we set up the framework:

- Random experiment:** Pick a random male customer and a random female customer, and record their haircut durations.

• **Mother variables:**

X	duration of a random man's haircut (in minutes)
Y	duration of a random woman's haircut (in minutes)

• **Other random variables:**

W_3	total duration of 3 random men's haircuts (in minutes)
$\bar{X}_3$	mean duration of 3 random men's haircuts (in minutes)
W_{5M}	total duration of 5 random men's haircuts (in minutes)
W_{2W}	total duration of 2 random women's haircuts (in minutes)
$D = W_{5M} - W_{2W}$	difference between the men's total and the women's total (in minutes)

• **Given:**

$\mu_X = 21$	population mean duration of all men's haircuts
$\sigma_X = 6$	population standard deviation of duration of all men's haircuts
$\mu_Y = 48$	population mean duration of all women's haircuts
$\sigma_Y = 12$	population standard deviation of duration of all women's haircuts

$$X \sim \mathcal{N}(\mu_X = 21, \sigma_X = 6)$$

$$Y \sim \mathcal{N}(\mu_Y = 48, \sigma_Y = 12)$$

all haircut durations are independent

.....
Note that 1.5 hours is 90 minutes. The duration of one man's haircut is normal. We're looking for the sum of 3 men's haircuts, so:

$$\left. \begin{array}{l} X_1, X_2, X_3 \sim \mathcal{N}(21, 6) \\ \text{independent} \end{array} \right\} \Rightarrow W_3 \sim \mathcal{N}(3 \cdot 21, \sqrt{3} \cdot 6) = \mathcal{N}(63, 10.3923)$$

Now we find the probability that it exceeds 90 minutes:

$$P(W_3 > 90) = \text{normalcdf}(90, E99, 63, 10.3923) \approx 0.0047$$

Answer: The probability that the total haircut time of 3 men exceeds 1.5 hours is 0.0047.

✓ **Solution 6: (b) Probability the mean duration of 3 men's haircuts is larger than 30 minutes**

This time we're looking for the mean instead of the sum, using the same 3 men's haircuts:

$$\left. \begin{array}{l} X_1, \dots, X_3 \sim \mathcal{N}(21, 6) \\ \text{independent} \end{array} \right\} \Rightarrow \bar{X}_3 \sim \mathcal{N}\left(21, \frac{6}{\sqrt{3}}\right) = \mathcal{N}(21, 3.4641)$$

Now we find the probability that it exceeds 30 minutes:

$$P(\bar{X}_3 > 30) = \text{normalcdf}(30, \text{E99}, 21, 3.4641) \approx 0.0047$$

This matches part (a) exactly - "total exceeds 90 minutes" and "mean exceeds 30 minutes" describe the exact same event, since $90/3 = 30$.

Answer: The probability that the mean haircut time for 3 men exceeds 30 minutes is 0.0047.

💡 **Tip: Comparing Two Variables**

To find a probability comparing two variables (e.g. $P(X > Y)$), move everything to one side first, turning it into a statement about their difference:

$$P(X > Y) = P(X - Y > 0)$$

Then find the distribution of $X - Y$ using the linear combination formula for two normal variables.

💡 Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✅ Solution 6: (c) Probability the total duration of 5 men's haircuts is smaller than that of 2 women's

Step 1: Find the distribution of the total duration of 5 men's haircuts.

The duration of one man's haircut is normal. We're looking for the sum of 5 men's haircuts, so:

$$\left. \begin{array}{l} X_1, \dots, X_5 \sim \mathcal{N}(21, 6) \\ \text{independent} \end{array} \right\} \Rightarrow W_{5M} \sim \mathcal{N}(5 \cdot 21, \sqrt{5} \cdot 6) = \mathcal{N}(105, 13.4164)$$

Step 2: Find the distribution of the total duration of 2 women's haircuts.

The duration of one woman's haircut is normal too. We're looking for the sum of 2 women's haircuts, so:

$$\left. \begin{array}{l} Y_1, Y_2 \sim \mathcal{N}(48, 12) \\ \text{independent} \end{array} \right\} \Rightarrow W_{2W} \sim \mathcal{N}(2 \cdot 48, \sqrt{2} \cdot 12) = \mathcal{N}(96, 16.9706)$$

Step 3: Find the distribution of the difference, and compute the probability.

We want $P(W_{5M} < W_{2W})$, so we rewrite it as a statement about the difference:

$$P(W_{5M} < W_{2W}) = P(W_{5M} - W_{2W} < 0)$$

Define $D = W_{5M} - W_{2W}$. This is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho = 0$ since the two sums are independent:

$$\mu_D = 105 - 96 = 9$$

$$\sigma_D = \sqrt{13.4164^2 + 16.9706^2} \approx 21.6333$$

So $D \sim \mathcal{N}(9, 21.6333)$. Now we find the probability:

$$P(D < 0) = \text{normalcdf}(-E99, 0, 9, 21.6333) \approx 0.3387$$

Answer: The probability that the total haircut time of 5 men is smaller than that of 2 women is 0.3387.

Question 7

38% of the Belgian people have blood group O^+ . Let W_{250} count the number of people with blood group O^+ in a group of 250 randomly selected Belgian people. What is the probability that there are at least 90 and less than 100 people with the blood group O^+ in a group of 250 randomly chosen Belgian people?

Give both an exact and an approximate calculation of this probability.

Tip: Binomial Distribution

Use the binomial distribution when a trial is repeated a fixed number of times, independently, with the same success probability each time:

$$Y \sim \text{Bin}(n, \pi), \quad P(Y = k) = \text{binompdf}(n, \pi, k)$$

Solution 7: (a) Exact calculation

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random Belgian person and record whether they have blood group O^+ .

- **Mother variable:**

$$U = \begin{cases} 1 & \text{if a random Belgian person has blood group } O^+ \\ 0 & \text{if a random Belgian person doesn't have blood group } O^+ \end{cases}$$

- **Other random variables:**

$$W_{250} \quad \text{number of the 250 random people that are } O^+$$

• **Given:**

$\pi = 0.38$ population proportion of all Belgian people with blood group O⁺
 $n = 250$ sample size

Step 1: Check the conditions for a binomial variable.

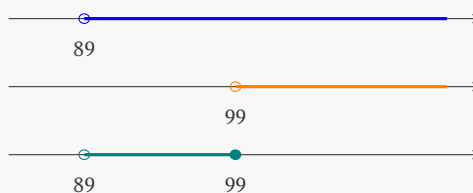
We check the 3 conditions:

- The underlying yes/no event is a person having blood group O⁺, with $\pi = 0.38$.
- It happens $n = 250$ times (one per person in the sample).
- The 250 people are independent of each other.

All 3 hold, so $W_{250} \sim \text{Bin}(250, 0.38)$.

Step 2: Translate the event into two one-sided pieces.

"At least 90 and less than 100" is the event $89 < W_{250} \leq 99$, i.e. $90 \leq W_{250} \leq 99$.



Looking at the colored regions above, we can express this as the difference between two regions:

$$P(90 \leq W_{250} \leq 99) = P(W_{250} > 89) - P(W_{250} > 99)$$

Step 3: Compute the probability.

Since `binomcdf` gives $P(W_{250} \leq k)$ directly, rewrite each piece in terms of that, then compute:

$$\begin{aligned} P(90 \leq W_{250} \leq 99) &= P(W_{250} \leq 99) - P(W_{250} \leq 89) \\ &= \text{binomcdf}(250, 0.38, 99) - \text{binomcdf}(250, 0.38, 89) \\ &\approx 0.7223 - 0.2376 = 0.4847 \end{aligned}$$

Answer: The exact probability is 0.4847.

💡 Tip: Normal Approximation to the Binomial

If $Y \sim \text{Bin}(n, \pi)$, we can approximate it using a normal distribution with matching mean and standard deviation:

$$Y \approx \mathcal{N}\left(n\pi, \sqrt{n\pi(1-\pi)}\right)$$

This approximation is only valid when **both** of the following hold:

- $n\pi \geq 5$
- $n(1-\pi) \geq 5$

✅ Solution 7: (b) Approximate calculation using the normal distribution

Step 1: Check the conditions.

Before approximating, we confirm both required conditions hold for $n = 250$, $\pi = 0.38$:

$$n\pi = 250 \cdot 0.38 = 95 \geq 5, \quad n(1-\pi) = 250 \cdot 0.62 = 155 \geq 5$$

Both hold, so the approximation is valid.

Step 2: Find the matching normal parameters.

The approximating normal distribution has the same mean and standard deviation as W_{250} :

$$\mu = n\pi = 95, \quad \sigma = \sqrt{n\pi(1-\pi)} = \sqrt{58.9} \approx 7.6746$$

Step 3: Approximate with `normalcdf`.

With μ and σ in hand, we compute the probability as if W_{250} were normal:

$$P(90 \leq W_{250} \leq 99) \approx \text{normalcdf}(90, 99, 95, 7.6746) \approx 0.4415$$

Answer: The approximate probability is 0.4415.

Question 8

The daily demand for a product in a shop can be modelled with the probability mass function:

k	0	1	2	3	4	5
$P(D = k)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

The demand during a day is independent of the demands during other days.

- What is the probability that the total demand during 365 days is at least 950?
- How many days do we have to wait in order to have at least 95% chance of observing a total demand that is at least 1800 units?
- What is the probability that we have to observe more than 40 days until the fifth day without demand for this product?

Tip: Mean of a Discrete Random Variable

Multiply each outcome by its own probability, then add:

$$\mu = E(X) = \sum_i x_i P(X=x_i)$$

Tip: Variance of a Discrete Random Variable - Convenience Formula

Do the same with each outcome squared to get $E(X^2)$, then subtract μ^2 :

$$\sigma^2 = E(X^2) - \mu^2, \quad \text{where } E(X^2) = \sum_i x_i^2 P(X=x_i)$$

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n} \sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 8: (a) Probability the total demand during 365 days is at least 950

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random day and record the demand for the product.
- **Mother variable:**

D demand for the product on a random day (in units)

- **Other random variables:**

W_n total demand during n random days (in units)

- **Given:** The probability distribution of D :

k	0	1	2	3	4	5
$P(D = k)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

The daily demands are independent across days.

.....

Step 1: Find the mean and standard deviation of a single day's demand.

Multiply each outcome by its probability, then add:

$$\mu = E(D) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{1}{8} + 2 \cdot \frac{2}{8} + 3 \cdot \frac{2}{8} + 4 \cdot \frac{1}{8} + 5 \cdot \frac{1}{8} = \frac{20}{8} = 2.5$$

Then use the convenience formula for the variance:

$$\begin{aligned}\sigma^2 &= E(D^2) - \mu^2 \\ &= \left(0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{1}{8} + 2^2 \cdot \frac{2}{8} + 3^2 \cdot \frac{2}{8} + 4^2 \cdot \frac{1}{8} + 5^2 \cdot \frac{1}{8} \right) - 2.5^2 \\ &= 8.5 - 6.25 = 2.25 \\ \sigma &= \sqrt{2.25} = 1.5\end{aligned}$$

Step 2: Apply the Central Limit Theorem to the total demand over 365 days.

The daily demand is not necessarily normal, so we check the conditions for the Central Limit Theorem:

$$\left. \begin{array}{l} D_1, \dots, D_{365} \text{ non normal} \\ \text{independent} \\ n = 365 \geq 30 \\ E(D) = 2.5, \sigma_D = 1.5 \end{array} \right\} \Rightarrow \begin{aligned} W_{365} &\approx \mathcal{N}(365 \cdot 2.5, \sqrt{365} \cdot 1.5) \\ &= \mathcal{N}(912.5, 28.658) \end{aligned}$$

Step 3: Compute the probability.

With W_{365} 's distribution in hand, we can find the probability it's at least 950:

$$P(W_{365} \geq 950) = \text{normalcdf}(950, E99, 912.5, 28.658) \approx 0.0953$$

Answer: The probability that the total demand is at least 950 is 0.0953.

Tip: Finding the Smallest n for a Probability Condition

When a question asks for the smallest sample size n satisfying a probability condition (e.g. $P(\dots) \geq 0.95$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the smallest X for which Y_1 satisfies the condition.

Solution 8: (b) Smallest number of days for at least 95% chance the total demand is at least 1800

Step 1: Generalize the distribution of the total demand to an arbitrary n .

Repeating part (a)'s reasoning with n left symbolic:

$$W_n \approx \mathcal{N}(2.5n, 1.5\sqrt{n})$$

Step 2: Search for the smallest n .

We want the smallest n such that:

$$P(W_n \geq 1800) \geq 0.95$$

We enter this into Y_1 :

$$Y_1 = \text{normalcdf}(1800, E99, 2.5X, 1.5\sqrt{X})$$

Searching around $n = 740$ using the table, we find that $n = 747$ is the smallest value for which the condition holds:

$$n = 747 \Rightarrow P(W_{747} \geq 1800) \approx 0.9502 \geq 0.95$$

Answer: The smallest number of days needed is 747.

Tip: Note: Negative Binomial Distribution Is Not Part of the Curriculum

This is an older-style exam question. **The negative binomial distribution (number of trials until the r -th success) is not part of this course's curriculum.** We solve it below anyway, using only the Binomial distribution.

Tip: Converting a "Trials Until the r -th Success" Question into a Binomial One

Let T_r be the number of trials needed to reach the r -th success. "The r -th success hasn't happened yet by trial k " is the same event as "fewer than r successes occurred among the first k trials," so:

$$P(T_r > k) = P(X_k \leq r - 1)$$

$$X_k \sim \text{Bin}(k, \pi)$$

✓ Solution 8: (c) Probability of observing more than 40 days until the 5th day without demand

Before computing anything, we extend the framework:

- **Other random variables:**

T_5	number of days observed until the 5th zero-demand day
X_{40}	number of zero-demand days among the first 40 days

Step 1: Set up, and convert to a Binomial count.

Let T_5 be the number of days until the 5th day with zero demand. A "success" here is a day with zero demand, $\pi = P(D = 0) = 1/8$. We want $P(T_5 > 40)$. "The 5th zero-demand day hasn't happened yet by day 40" is the same event as "fewer than 5 zero-demand days occurred among the first 40 days":

$$P(T_5 > 40) = P(X_{40} \leq 4)$$

X_{40} is binomial, since each of the first 40 days independently has zero demand with probability $\pi = 1/8$:

$$X_{40} \sim \text{Bin}(40, 1/8)$$

Step 2: Compute.

Find the binomial probability on the calculator:

$$P(T_5 > 40) = P(X_{40} \leq 4) = \text{binomcdf}(40, 1/8, 4) \approx 0.4287$$

Answer: $P(T_5 > 40) \approx 0.4287$.

Question 9

The duration of a random pregnancy is a left-skewed continuous random variable with expectation 280 days and standard deviation 12 days. What is the probability that the average duration of 50 pregnancies is at most 275 days?

Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

Solution 9:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random pregnancy and record its duration.
- **Mother variable:**

X duration of a random pregnancy (in days)

• **Other random variables:**

$\bar{X}_{50}$ mean duration of 50 random pregnancies (in days)

• **Given:**

$\mu = 280$ population mean duration of all pregnancies

$\sigma = 12$ population standard deviation of duration of all pregnancies

$n = 50$ sample size

X is left-skewed

.....

Step 1: Apply the Central Limit Theorem.

Even though a single pregnancy's duration is left-skewed, it's non normal, but $n = 50$ is large enough to apply the Central Limit Theorem:

$$\left. \begin{array}{l} X_1, \dots, X_{50} \text{ non normal} \\ \text{independent} \\ n = 50 \geq 30 \end{array} \right\} \Rightarrow \bar{X}_{50} \approx \mathcal{N}\left(280, \frac{12}{\sqrt{50}}\right) = \mathcal{N}(280, 1.6971)$$

Step 2: Compute the probability.

With $\bar{X}_{50}$'s distribution in hand, we find the probability it's at most 275 days:

$$P(\bar{X}_{50} \leq 275) = \text{normalcdf}(-E99, 275, 280, 1.6971) \approx 0.0016$$

Answer: The probability that the mean duration of 50 pregnancies is at most 275 days is 0.0016.

Question 10

The body lengths of students are normally distributed with (population) mean 178.6 cm and (population) standard deviation 7.6 cm.

- What is the probability that a randomly chosen student is taller than 1.9 m? Interpret this probability in the context of the population.
- What is the probability that the average body length of two randomly chosen students is larger than 1.9 m?
- What is the probability that the average body length of ten randomly chosen students is larger than 1.9 m?
- Graphically display the probabilities that you have calculated in a-c.

✓ Solution 10: (a) Probability a random student is taller than 1.9 m, and its interpretation

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random student and record their body length.

- **Mother variable:**

X body length of a random student (in cm)

- **Given:**

$\mu = 178.6$ population mean body length of all students

$\sigma = 7.6$ population standard deviation of body length of all students

$X \sim \mathcal{N}(\mu = 178.6, \sigma = 7.6)$

.....

1.9 m is 190 cm. Using the population distribution directly:

$$P(X > 190) = \text{normalcdf}(190, \text{E99}, 178.6, 7.6) \approx 0.0668$$

Answer: The probability is 0.0668 - we expect about 6.68% of all students to be taller than 1.9 m.

💡 Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

✅ Solution 10: (b) Probability the mean body length of 2 students is larger than 1.9 m

Step 1: Find the distribution of $\bar{X}_2$.

The body length of one student is normal. We're looking for the mean of 2 students, so:

$$\left. \begin{array}{l} X_1, X_2 \sim \mathcal{N}(178.6, 7.6) \\ \text{independent} \end{array} \right\} \Rightarrow \bar{X}_2 \sim \mathcal{N}\left(178.6, \frac{7.6}{\sqrt{2}}\right) = \mathcal{N}(178.6, 5.3740)$$

Step 2: Compute the probability.

Now that we know the distribution of $\bar{X}_2$, we can find the probability it's larger than 190:

$$P(\bar{X}_2 > 190) = \text{normalcdf}(190, \text{E99}, 178.6, 5.3740) \approx 0.01695$$

Answer: The probability that the mean body length of two students is larger than 1.9 m is 0.01695.

✓ **Solution 10: (c) Probability the mean body length of 10 students is larger than 1.9 m**

Step 1: Find the distribution of $\bar{X}_{10}$.

The same reasoning as part (b) applies, now with $n = 10$:

$$\left. \begin{array}{l} X_1, \dots, X_{10} \sim \mathcal{N}(178.6, 7.6) \\ \text{independent} \end{array} \right\} \Rightarrow \bar{X}_{10} \sim \mathcal{N}\left(178.6, \frac{7.6}{\sqrt{10}}\right) = \mathcal{N}(178.6, 2.4033)$$

Step 2: Compute the probability.

Now that we know the distribution of $\bar{X}_{10}$, we can find the probability it's larger than 190:

$$P(\bar{X}_{10} > 190) = \text{normalcdf}(190, \text{E99}, 178.6, 2.4033) \approx 1.0507 \cdot 10^{-6}$$

Answer: The probability that the mean body length of ten students is larger than 1.9 m is $1.0507 \cdot 10^{-6}$, extremely small.

✓ **Solution 10: (d) Graphically display the probabilities that you have calculated in a-c.**

To visualize the results, we plot the three probability density functions for the distributions corresponding to $n = 1$, $n = 2$, and $n = 10$.

- For $n = 1$, the distribution is $X \sim N(178.6, 7.6)$, entered in the TI-84 as $Y1 = \text{normalpdf}(X, 178.6, 7.6)$.

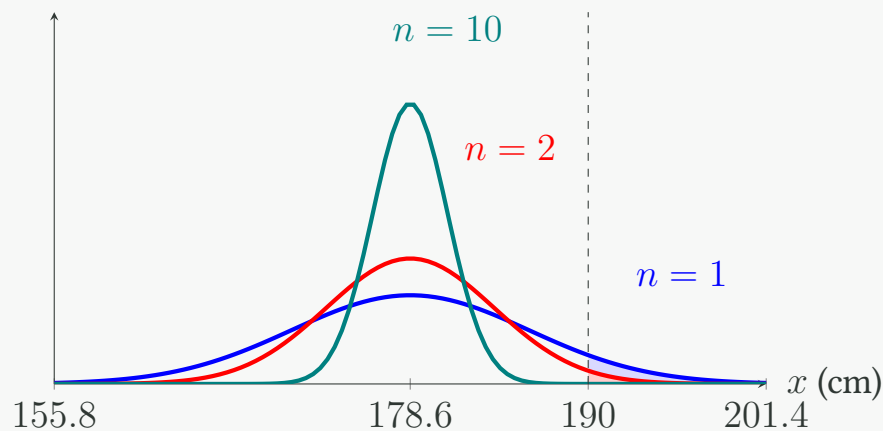
- For $n = 2$, the distribution is $\bar{X}_2 \sim N(178.6, 7.6/\sqrt{2})$, entered as
`Y2 = normalpdf(X, 178.6, 7.6/√2).`
- For $n = 10$, the distribution is $\bar{X}_{10} \sim N(178.6, 7.6/\sqrt{10})$, entered as
`Y3 = normalpdf(X, 178.6, 7.6/√10).`

We set the viewing window. A good range is three standard deviations around the mean:

$$[\mu - 3\sigma, \mu + 3\sigma] = [178.6 - 3(7.6), 178.6 + 3(7.6)] = [155.8, 201.4]$$

Finally, we shade the area for $x > 190$ under each curve using the calculator's integral function, found by pressing `2nd + TRACE → 7: ∫ f(x) dx`.

Answer: The graph shows that as the sample size n increases, the distribution of the sample mean becomes narrower, and the probability of an extreme mean decreases.



Question 11

The weight (in grams) of a randomly chosen butter dish from a factory is normally distributed with a population mean of μ and a population standard deviation of $\sigma = 3$ grams. An inspector takes a random sample of 25 butter dishes and measures their weights. If the sample mean weight of these 25 dishes is less than 250 grams, the factory has to pay a fine.

- What is the probability that the factory receives a fine if the true population mean is $\mu = 251$ grams?
- Determine the value for the population mean μ such that the probability of receiving a fine is exactly 1%.

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Solution 11: (a) Probability of a fine when $\mu = 251$ grams

Before computing anything, we set up the framework:

- Random experiment:** Pick a random butter dish and record its weight.
- Mother variable:**

X weight of a random butter dish (in g)

- **Other random variables:**

$\bar{X}_{25}$ mean weight of 25 random butter dishes (in g)

- **Given:**

$\mu = ?$ population mean weight of all butter dishes

$\sigma = 3$ population standard deviation of weight of all butter dishes

$n = 25$ sample size

$X \sim \mathcal{N}(\mu = ?, \sigma = 3)$

a fine is issued if the observed sample mean is below 250 g

.....

Step 1: Find the distribution of $\bar{X}_{25}$, assuming $\mu = 251$.

The weight of one dish is normal. We're looking for the mean of 25 dishes, so:

$$\left. \begin{array}{l} X_1, \dots, X_{25} \sim \mathcal{N}(251, 3) \\ \text{independent} \end{array} \right\} \Rightarrow \bar{X}_{25} \sim \mathcal{N}\left(251, \frac{3}{\sqrt{25}}\right) = \mathcal{N}(251, 0.6)$$

Step 2: Compute the probability.

Now that we know the distribution of $\bar{X}_{25}$, we can find the probability it's below 250:

$$P(\bar{X}_{25} < 250) = \text{normalcdf}(-E99, 250, 251, 0.6) \approx 0.0478$$

Answer: The probability of a fine is 0.0478.

Tip: Solving for a Missing μ or σ with the Numeric Solver

When a distribution for W_n or $\bar{X}_n$ contains an unknown μ or σ , and a probability statement is given instead:

1. Write the probability statement as a calculator equation, with the unknown left as a variable, e.g. $\text{normalcdf}(-E99, 250, X, 0.6) = 0.01$ (μ unknown).
2. Open the numeric solver (MATH $\rightarrow$ Solver . . .). Let the calculator's variable X represent the actual unknown (μ or σ) - not the same X as the mother variable, just a naming coincidence of the calculator's interface.
3. Set E1: the full normalcdf expression with X in place of the unknown, and E2: the target probability.
4. Enter an initial guess for X, then press ALPHA + ENTER (SOLVE).

Solution 11: (b) Value of μ such that the probability of a fine is exactly 1%

Step 1: Set up the equation.

We want $P(\bar{X}_{25} < 250) = 0.01$, with μ unknown:

$$\text{normalcdf}(-E99, 250, X, 0.6) = 0.01$$

Step 2: Solve using the numeric solver.

Set E1: $\text{normalcdf}(-E99, 250, X, 0.6)$, E2: 0.01, an initial guess $X = 250$, then SOLVE:

$$\mu \approx 251.3956$$

Answer: The mean μ should be 251.40 grams.

Question 12

A physician leaves her home to drive by car to the hospital, examines 5 patients in the hospital, and returns home. The time to drive from her home to the hospital and the time to drive from the hospital to her home are both normally distributed with expectation 10 min and standard deviation 2 min. The time to examine each patient is normally distributed with expectation 12 min and standard deviation 3 min. Assume that all these times are independent.

What is the probability that more than 75 min elapse between the physician's departure from her home and her arrival at her home?

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ Solution 12:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random driving trip and a random patient examination, and record their durations.

- **Mother variables:**

X duration of a random one-way driving trip (in minutes)

Y duration of a random patient examination (in minutes)

- **Other random variables:**

W_X total driving time for the two one-way trips (in minutes)

W_Y total examination time for the 5 patients (in minutes)

$T = W_X + W_Y$ total elapsed time from departure to return (in minutes)

- **Given:**

$\mu_X = 10$ population mean duration of all driving trips

$\sigma_X = 2$ population standard deviation of duration of all driving trips

$\mu_Y = 12$ population mean duration of all patient examinations

$\sigma_Y = 3$ population standard deviation of duration of all patient examinations

$X \sim \mathcal{N}(\mu_X = 10, \sigma_X = 2)$

$Y \sim \mathcal{N}(\mu_Y = 12, \sigma_Y = 3)$

all durations are independent

Step 1: Find the distributions of the total driving time and total examination time.

The physician drives twice (there and back) and examines 5 patients. The driving time is normal,

so:

$$\left. \begin{array}{l} X_1, X_2 \sim \mathcal{N}(10, 2) \\ \text{independent} \end{array} \right\} \Rightarrow W_X \sim \mathcal{N}(2 \cdot 10, \sqrt{2} \cdot 2) = \mathcal{N}(20, 2.8284)$$

The examination time is normal too, so:

$$\left. \begin{array}{l} Y_1, \dots, Y_5 \sim \mathcal{N}(12, 3) \\ \text{independent} \end{array} \right\} \Rightarrow W_Y \sim \mathcal{N}(5 \cdot 12, \sqrt{5} \cdot 3) = \mathcal{N}(60, 6.7082)$$

Step 2: Find the distribution of the total elapsed time.

The total elapsed time is $T = W_X + W_Y$, a linear combination with $a = 1$, $b = 1$, $c = 0$, and $\rho = 0$ since the driving and examination times are independent:

$$\mu_T = 20 + 60 = 80$$

$$\sigma_T = \sqrt{2.8284^2 + 6.7082^2} \approx 7.2801$$

So $T \sim \mathcal{N}(80, 7.2801)$.

Step 3: Compute the probability.

With T 's distribution in hand, we find the probability it exceeds 75:

$$P(T > 75) = \text{normalcdf}(75, \text{E99}, 80, 7.2801) \approx 0.7539$$

Answer: The probability that more than 75 minutes elapse is 0.7539.

Question 13

An exam consists of 40 questions. The time that a random student needs to answer each question is uniformly distributed on the interval $[4, 8]$ minutes. Assume that the time to answer a question is independent of the time to answer other questions.

What is the probability that the student finishes his/her exam in less than 230 minutes?

Tip: Continuous Uniform Distribution

Whenever a value is picked at random from an interval $[a, b]$, with no value in it more likely than any other, use the **continuous uniform distribution**:

$$X \sim \mathcal{U}(a, b)$$

Tip: Mean of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **mean** is:

$$\mu = \int_a^b x \cdot f(x) dx = \text{fnInt}(x \cdot f(x), x, a, b)$$

We never compute this integral by hand - we always use `fnInt` on the calculator.

Tip: Variance of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **variance** is:

$$\sigma^2 = \int_a^b x^2 \cdot f(x) dx - \mu^2 = \text{fnInt}(x^2 \cdot f(x), x, a, b) - \mu^2$$

Just like the mean, we never compute this integral by hand - we always use `fnInt` on the calculator.

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n} \sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 13:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random exam question and record the time to answer it.
- **Mother variable:**

X time to answer a random exam question (in minutes)

- **Other random variables:**

W_{40} total time to answer 40 random exam questions (in minutes)

• **Given:**

$a = 4$ lower bound of the (uniform) time to answer a question
 $b = 8$ upper bound of the (uniform) time to answer a question

$$X \sim \mathcal{U}(4, 8)$$

the 40 questions' answer times are independent

.....
Step 1: Find the density of X .

The time to answer each question is picked at random from $[4, 8]$, with no value more likely than any other - this is the continuous uniform distribution. The density is flat over $[4, 8]$:

$$f(x) = \frac{1}{8 - 4} = \frac{1}{4}, \quad 4 \leq x \leq 8$$

Step 2: Find the mean and standard deviation of a single question's time.

Integrate $x \cdot f(x)$ over the domain:

$$\mu = \text{fnInt}(X \cdot (1/4), X, 4, 8) = 6$$

Then integrate $x^2 \cdot f(x)$ and subtract μ^2 :

$$\sigma^2 = \text{fnInt}(X^2 \cdot (1/4), X, 4, 8) - 6^2 = \frac{112}{3} - 36 = \frac{4}{3}$$

$$\sigma = \sqrt{4/3} \approx 1.1547$$

Step 3: Apply the Central Limit Theorem to the total time over 40 questions.

The time per question is not necessarily normal (it's uniform), so we check the conditions for

the Central Limit Theorem:

$$\left. \begin{array}{l} X_1, \dots, X_{40} \text{ non normal} \\ \text{independent} \\ n = 40 \geq 30 \\ E(X) = 6, \sigma_X = 1.1547 \end{array} \right\} \Rightarrow \begin{array}{l} W_{40} \approx \mathcal{N}(40 \cdot 6, \sqrt{40} \cdot 1.1547) \\ = \mathcal{N}(240, 7.3030) \end{array}$$

Step 4: Compute the probability.

With W_{40} 's distribution in hand, we find the probability it's less than 230:

$$P(W_{40} < 230) = \text{normalcdf}(-E99, 230, 240, 7.3030) \approx 0.0855$$

Answer: The probability that the student finishes in less than 230 minutes is 0.0855.

Question 14

A bicycle chain is composed of links. The length of a single link is a random variable with a population mean of $\mu = 0.5$ cm and a population standard deviation of $\sigma = 0.04$ cm. The lengths of different links are independent and identically distributed. For a chain to be acceptable, its total length must be between 49 cm and 50 cm.

- What is the proportion of chains with 100 links that meet this length requirement?
- What is the proportion of chains with 99 links that meet this length requirement?
- Assume that bicycle chains are made with 99 links and that the population standard deviation σ of a single link's length can be adjusted. What should be the value of σ such that at least 90% of the bicycle chains meet the length requirement?

Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 14: (a) Proportion of chains with 100 links that meet the length requirement

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random link and record its length.
- **Mother variable:**

X length of a random link (in cm)

- **Other random variables:**

W_n total length of a chain with n random links (in cm)

- **Given:**

$\mu = 0.5$ population mean length of all links

$\sigma = 0.04$ population standard deviation of length of all links

link lengths are independent

acceptable if total length is between 49 and 50 cm

Step 1: Apply the Central Limit Theorem for $n = 100$ links.

The length of one link is not necessarily normal, so we check the conditions for the Central Limit Theorem:

$$\left. \begin{array}{l} X_1, \dots, X_{100} \text{ non normal} \\ \text{independent} \\ n = 100 \geq 30 \\ E(X) = 0.5, \sigma_X = 0.04 \end{array} \right\} \Rightarrow \begin{array}{l} W_{100} \approx \mathcal{N}(100 \cdot 0.5, \sqrt{100} \cdot 0.04) \\ = \mathcal{N}(50, 0.4) \end{array}$$

Step 2: Compute the probability.

With W_{100} 's distribution in hand, we find the probability it falls between 49 and 50:

$$P(49 \leq W_{100} \leq 50) = \text{normalcdf}(49, 50, 50, 0.4) \approx 0.4938$$

Answer: 49.38% of chains with 100 links meet the requirement.

✓ Solution 14: (b) Proportion of chains with 99 links that meet the length requirement

Step 1: Apply the Central Limit Theorem for $n = 99$ links.

The same reasoning as part (a) applies, now with $n = 99$:

$$\left. \begin{array}{l} X_1, \dots, X_{99} \text{ non normal} \\ \text{independent} \\ n = 99 \geq 30 \\ E(X) = 0.5, \sigma_X = 0.04 \end{array} \right\} \Rightarrow \begin{array}{l} W_{99} \approx \mathcal{N}(99 \cdot 0.5, \sqrt{99} \cdot 0.04) \\ = \mathcal{N}(49.5, 0.3980) \end{array}$$

Step 2: Compute the probability.

With W_{99} 's distribution in hand, we find the probability it falls between 49 and 50:

$$P(49 \leq W_{99} \leq 50) = \text{normalcdf}(49, 50, 49.5, 0.3980) \approx 0.7910$$

Answer: 79.10% of chains with 99 links meet the requirement.

Tip: Solving for a Missing μ or σ with the Numeric Solver

When a distribution for W_n or $\bar{X}_n$ contains an unknown μ or σ , and a probability statement is given instead:

1. Write the probability statement as a calculator equation, with the unknown left as a variable, e.g. $\text{normalcdf}(-E99, 250, X, 0.6) = 0.01$ (μ unknown).
2. Open the numeric solver (MATH $\rightarrow$ Solver . . .). Let the calculator's variable X represent the actual unknown (μ or σ) - not the same X as the mother variable, just a naming coincidence of the calculator's interface.
3. Set E1: the full normalcdf expression with X in place of the unknown, and E2: the target probability.
4. Enter an initial guess for X, then press ALPHA + ENTER (SOLVE).

Solution 14: (c) Value of σ so at least 90% of 99-link chains meet the requirement

Step 1: Set up the equation.

We want $P(49 \leq W_{99} \leq 50) = 0.90$, where $W_{99} \approx \mathcal{N}(49.5, \sqrt{99}\sigma)$ and σ is unknown:

$$\text{normalcdf}(49, 50, 49.5, \sqrt{99} \cdot X) = 0.90$$

Step 2: Solve using the numeric solver.

Set E1: $\text{normalcdf}(49, 50, 49.5, \sqrt{99} \cdot X)$, E2: 0.90, an initial guess $X = 0.04$, then SOLVE:

$$\sigma \approx 0.0306$$

Answer: The standard deviation σ should be 0.0306 cm.

Question 15

In a manuscript of 100 pages, the number of typos on each page has the following probability mass function:

k	0	1	2	3
$P(X = k)$	0.6	0.25	0.1	0.05

Assume that the number of typos on a page is independent of the number of typos on other pages. What is the probability that the manuscript contains at least 70 typos?

Tip: Mean of a Discrete Random Variable

Multiply each outcome by its own probability, then add:

$$\mu = E(X) = \sum_i x_i P(X=x_i)$$

Tip: Variance of a Discrete Random Variable - Convenience Formula

Do the same with each outcome squared to get $E(X^2)$, then subtract μ^2 :

$$\sigma^2 = E(X^2) - \mu^2, \quad \text{where } E(X^2) = \sum_i x_i^2 P(X=x_i)$$

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n} \sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✔ Solution 15:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random page from the manuscript and record its number of typos.
- **Mother variable:**

X number of typos on a random page

- **Other random variables:**

W_{100} total number of typos in the 100-page manuscript

- **Given:** The probability distribution of X :

k	0	1	2	3
$P(X = k)$	0.6	0.25	0.1	0.05

The number of typos on each page is independent across pages.

Step 1: Find the mean and standard deviation of a single page's typos.

Multiply each outcome by its probability, then add:

$$\mu = E(X) = 0 \cdot 0.6 + 1 \cdot 0.25 + 2 \cdot 0.1 + 3 \cdot 0.05 = 0.6$$

Then use the convenience formula for the variance:

$$\sigma^2 = E(X^2) - \mu^2 = (0^2 \cdot 0.6 + 1^2 \cdot 0.25 + 2^2 \cdot 0.1 + 3^2 \cdot 0.05) - 0.6^2 = 1.1 - 0.36 = 0.74$$

$$\sigma = \sqrt{0.74} \approx 0.8602$$

Step 2: Apply the Central Limit Theorem to the total typos over 100 pages.

The number of typos on a page is not necessarily normal, so we check the conditions for the Central Limit Theorem:

$$\left. \begin{array}{l} X_1, \dots, X_{100} \text{ non normal} \\ \text{independent} \\ n = 100 \geq 30 \\ E(X) = 0.6, \sigma_X = 0.8602 \end{array} \right\} \Rightarrow \begin{array}{l} W_{100} \approx \mathcal{N}(100 \cdot 0.6, \sqrt{100} \cdot 0.8602) \\ = \mathcal{N}(60, 8.6023) \end{array}$$

Step 3: Compute the probability.

With W_{100} 's distribution in hand, we find the probability it's at least 70:

$$P(W_{100} \geq 70) \approx \text{normalcdf}(70, E99, 60, 8.6023) \approx 0.1225$$

Answer: The probability that the manuscript contains at least 70 typos is 0.1225.

Question 16

The bill amounts in a snack bar have a mean of €5 and a standard deviation of €2.

- What is the probability that the total revenue is at least €375 on a day with 64 customers (if each customer pays a bill)?
- How many customers are needed in order for the total revenue to exceed €500 with at least 95% probability?

Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 16: (a) Probability the total revenue is at least €375 with 64 customers

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random customer at the snack bar and record their bill amount.

- **Mother variable:**

X bill amount of a random customer (in €)

- **Other random variables:**

W_n total revenue from n random customers (in €)

- **Given:**

$\mu = 5$ population mean bill amount of all customers

$\sigma = 2$ population standard deviation of bill amount of all customers

bill amounts are independent and identically distributed

Step 1: Apply the Central Limit Theorem for $n = 64$ customers.

A single bill amount is not necessarily normal, so we check the conditions for the Central Limit Theorem:

$$\left. \begin{array}{l} X_1, \dots, X_{64} \text{ non normal} \\ \text{independent} \\ n = 64 \geq 30 \\ E(X) = 5, \sigma_X = 2 \end{array} \right\} \Rightarrow \begin{array}{l} W_{64} \approx \mathcal{N}(64 \cdot 5, \sqrt{64} \cdot 2) \\ = \mathcal{N}(320, 16) \end{array}$$

Step 2: Compute the probability.

With W_{64} 's distribution in hand, we find the probability it's at least 375:

$$P(W_{64} \geq 375) = \text{normalcdf}(375, E99, 320, 16) \approx 0.000294$$

Answer: The probability that the total revenue is at least €375 is 0.000294.

Tip: Finding the Smallest n for a Probability Condition

When a question asks for the smallest sample size n satisfying a probability condition (e.g. $P(\dots) \geq 0.95$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the smallest X for which Y_1 satisfies the condition.

Solution 16: (b) Smallest number of customers for the revenue to exceed €500 with 95% probability

Step 1: Generalize the distribution of the total revenue to an arbitrary n .

Repeating part (a)'s reasoning with n left symbolic:

$$W_n \approx \mathcal{N}(5n, 2\sqrt{n})$$

Step 2: Search for the smallest n .

We want the smallest n such that:

$$P(W_n > 500) \geq 0.95$$

We enter this into Y_1 :

$$Y_1 = \text{normalcdf}(500, E99, 5X, 2\sqrt{X})$$

Searching around $n = 100$ using the table, we find that $n = 107$ is the smallest value for which the condition holds:

$$n = 107 \quad \Rightarrow \quad P(W_{107} > 500) \approx 0.9547 \geq 0.95$$

Answer: At least 107 customers are needed.

Question 17

The fuel consumption of "Light," a new type of car built by Biocar, is normally distributed with a population standard deviation of $\sigma = 0.5$ litres per 100 km. Biocar claims that the population mean fuel consumption, μ , does not exceed 3 litres per 100 km.

A consumers' organisation wants to test this claim and takes a random sample of $n = 25$ "Light" cars. They find that the sample mean fuel consumption is $\bar{x} = 3.3$ litres per 100 km.

Assume that the true population mean fuel consumption is exactly $\mu = 3$ litres per 100 km (the value at the boundary of Biocar's claim). What is the probability of observing a sample mean of at least 3.3 litres per 100 km? (This probability is known as a p-value in hypothesis testing.)

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Solution 17:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random "Light" car and record its fuel consumption.
- **Mother variable:**

X fuel consumption of a random "Light" car (in litres per 100 km)

• **Other random variables:**

$\bar{X}_{25}$ mean fuel consumption of 25 random "Light" cars (in litres per 100 km)

• **Given:**

$\mu = 3$ population mean fuel consumption, assumed exactly at the boundary of Biocar's claim

$\sigma = 0.5$ population standard deviation of fuel consumption of all "Light" cars

$n = 25$ sample size

$\bar{x}_{25} = 3.3$ observed sample mean fuel consumption of 25 randomly chosen "Light" cars

$$X \sim \mathcal{N}(\mu = 3, \sigma = 0.5)$$

.....

Step 1: Find the distribution of the sample mean, assuming $\mu = 3$.

The fuel consumption of one car is normal. We're looking for the mean of 25 cars, so:

$$\left. \begin{array}{l} X_1, \dots, X_{25} \sim \mathcal{N}(3, 0.5) \\ \text{independent} \end{array} \right\} \Rightarrow \bar{X}_{25} \sim \mathcal{N}\left(3, \frac{0.5}{\sqrt{25}}\right) = \mathcal{N}(3, 0.1)$$

Step 2: Compute the probability (the p-value).

With $\bar{X}_{25}$'s distribution in hand, we find the probability it's at least 3.3:

$$P(\bar{X}_{25} \geq 3.3) = \text{normalcdf}(3.3, \text{E99}, 3, 0.1) \approx 0.00135$$

Answer: The probability of observing a sample mean of at least 3.3 litres per 100 km is 0.00135. This very low probability (p-value) suggests it is highly unlikely to observe a sample mean of 3.3 if the true mean were actually 3.

Question 18

The weights of packages of coffee are normally distributed with mean 250 g and standard deviation 3 g.

- We randomly choose 16 packages and, afterwards, a seventeenth package. What is the probability that the seventeenth package weighs at least 2 g more than the average weight of the first sixteen packages?
- We take two random samples of packages, both of size 64. What is the probability that the difference between the average weights of these samples is larger than 1 g?

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ **Solution 18: (a) Probability the 17th package weighs at least 2 g more than the mean of the first 16**

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random package of coffee and record its weight.
- **Mother variable:**

X weight of a random package of coffee (in g)

- **Other random variables:**

$\bar{X}_{16}$ mean weight of the first 16 random packages (in g)

$D = X_{17} - \bar{X}_{16}$ difference between the 17th package's weight and the mean of the first 16 (in g)

- **Given:**

$\mu = 250$ population mean weight of all packages

$\sigma = 3$ population standard deviation of weight of all packages

$X \sim \mathcal{N}(\mu = 250, \sigma = 3)$

package weights are independent

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Step 1: Find the distribution of $\bar{X}_{16}$.

The weight of one package is normal. We're looking for the mean of 16 packages, so:

$$\left. \begin{array}{l} X_1, \dots, X_{16} \sim \mathcal{N}(250, 3) \\ \text{independent} \end{array} \right\} \Rightarrow \bar{X}_{16} \sim \mathcal{N}\left(250, \frac{3}{\sqrt{16}}\right) = \mathcal{N}(250, 0.75)$$

Step 2: Find the distribution of the difference $D = X_{17} - \bar{X}_{16}$.

X_{17} and $\bar{X}_{16}$ are independent normal variables, so this is a linear combination with $a = 1$,

$b = -1, c = 0$:

$$\mu_D = 250 - 250 = 0$$

$$\sigma_D = \sqrt{3^2 + 0.75^2} \approx 3.0923$$

So $D \sim \mathcal{N}(0, 3.0923)$.

Step 3: Compute the probability.

With D 's distribution in hand, we find the probability it's at least 2:

$$P(D \geq 2) = \text{normalcdf}(2, \text{E99}, 0, 3.0923) \approx 0.2589$$

Answer: The probability is 0.2589.

✓ Solution 18: (b) Probability the difference between two sample means (each $n = 64$) is larger than 1 g

Before computing anything, we extend the framework:

• Other random variables:

$\bar{X}_A$

mean weight of a random sample of 64 packages

$\bar{X}_B$

mean weight of a second, independent random sample of 64 packages

$D = \bar{X}_A - \bar{X}_B$

difference between the two samples' mean weights (in g)

Step 1: Find the distribution of a single sample mean.

The weight of one package is normal, so:

$$\left. \begin{array}{l} X_1, \dots, X_{64} \sim \mathcal{N}(250, 3) \\ \text{independent} \end{array} \right\} \Rightarrow \bar{X}_{64} \sim \mathcal{N}\left(250, \frac{3}{\sqrt{64}}\right) = \mathcal{N}(250, 0.375)$$

This applies to both $\bar{X}_A$ and $\bar{X}_B$.

Step 2: Translate the event into two one-sided pieces.

"Differs by more than 1 g" doesn't say which sample has the bigger mean - either $\bar{X}_A$ could exceed $\bar{X}_B$ by more than 1 g, or $\bar{X}_B$ could exceed $\bar{X}_A$ by more than 1 g. These are two separate, non-overlapping events, so we add their probabilities instead of using absolute value notation:

$$P(D > 1) + P(D < -1), \quad D = \bar{X}_A - \bar{X}_B$$

Step 3: Find the distribution of the difference.

This is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho = 0$ since the two samples are independent:

$$\mu_D = 250 - 250 = 0$$

$$\sigma_D = \sqrt{0.375^2 + 0.375^2} \approx 0.5303$$

So $D \sim \mathcal{N}(0, 0.5303)$.

Step 4: Compute the probability.

By symmetry, both tails carry the same probability:

$$P(D > 1) = \text{normalcdf}(1, \text{E99}, 0, 0.5303) \approx 0.0297$$

$$P(D < -1) = \text{normalcdf}(-\text{E99}, -1, 0, 0.5303) \approx 0.0297$$

Adding them:

$$P(D > 1) + P(D < -1) \approx 0.0297 + 0.0297 = 0.0593$$

Answer: The probability that the sample means differ by more than 1 g is 0.0593.

Question 19

The lifespans of batteries of brand A are normally distributed with mean 3600 hours and standard deviation 300 hours. The lifespans of batteries of brand B are normally distributed with mean 3700 hours and standard deviation 400 hours.

- What is the probability that the mean lifespan of a random sample of batteries from brand A is larger than the mean lifespan of a random sample of batteries from brand B, if both sample sizes are equal to 25?
- In the case of equal sample sizes, what should be this sample size in order for the probability in (a) to be smaller than 0.01?

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ **Solution 19: (a) Probability the mean lifespan of 25 brand A batteries exceeds that of 25 brand B batteries**

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random brand A battery and a random brand B battery, and record their lifespans.

- **Mother variables:**

X lifespan of a random brand A battery (in hours)

Y lifespan of a random brand B battery (in hours)

- **Other random variables:**

$\bar{X}_n$ mean lifespan of a random sample of n brand A batteries (in hours)

$\bar{Y}_n$ mean lifespan of a random sample of n brand B batteries (in hours)

$D_n = \bar{X}_n - \bar{Y}_n$ difference between the two samples' mean lifespans (in hours)

- **Given:**

$\mu_X = 3600$ population mean lifespan of all brand A batteries

$\sigma_X = 300$ population standard deviation of lifespan of all brand A batteries

$\mu_Y = 3700$ population mean lifespan of all brand B batteries

$\sigma_Y = 400$ population standard deviation of lifespan of all brand B batteries

$X \sim \mathcal{N}(\mu_X = 3600, \sigma_X = 300)$

$Y \sim \mathcal{N}(\mu_Y = 3700, \sigma_Y = 400)$

all battery lifespans are independent

.....

Step 1: Find the distribution of each sample mean.

Let $\bar{X}_{25}$ and $\bar{Y}_{25}$ be the means of two independent samples, each of size $n = 25$. Both lifespans are normal, so:

$$\bar{X}_{25} \sim \mathcal{N}\left(3600, \frac{300}{\sqrt{25}}\right) = \mathcal{N}(3600, 60), \quad \bar{Y}_{25} \sim \mathcal{N}\left(3700, \frac{400}{\sqrt{25}}\right) = \mathcal{N}(3700, 80)$$

Step 2: Find the distribution of the difference $D = \bar{X}_{25} - \bar{Y}_{25}$.

This is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho = 0$ since the two samples are independent:

$$\mu_D = 3600 - 3700 = -100$$

$$\sigma_D = \sqrt{60^2 + 80^2} = 100$$

So $D \sim \mathcal{N}(-100, 100)$.

Step 3: Compute the probability.

We want $P(\bar{X}_{25} > \bar{Y}_{25}) = P(D > 0)$:

$$P(D > 0) = \text{normalcdf}(0, \text{E99}, -100, 100) \approx 0.1587$$

Answer: The probability is 0.1587.

 Tip: Finding the Smallest n for a Probability Condition

When a question asks for the smallest sample size n satisfying a probability condition (e.g. $P(\dots) < 0.01$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the smallest X for which Y_1 satisfies the con-

dition.

✓ **Solution 19: (b) Sample size so the probability in (a) is smaller than 0.01**

Step 1: Generalize the distribution of the difference to an arbitrary n .

Repeating part (a)'s reasoning with both sample sizes left symbolic (and equal):

$$D_n \sim \mathcal{N}\left(-100, \sqrt{\frac{300^2}{n} + \frac{400^2}{n}}\right) = \mathcal{N}\left(-100, \frac{500}{\sqrt{n}}\right)$$

Step 2: Search for the smallest n .

We want the smallest n such that:

$$P(D_n > 0) < 0.01$$

We enter this into Y_1 :

$$Y_1 = \text{normalcdf}(0, E99, -100, 500/\sqrt{X})$$

Searching around $n = 130$ using the table, we find that $n = 136$ is the smallest value for which the condition holds:

$$n = 136 \Rightarrow P(D_{136} > 0) \approx 0.0098 < 0.01$$

Answer: The sample size should be at least 136.

Question 20

The cargo carrying capacity of a passenger plane is 10 tons. The random pair (X, Y) measures the weight of the (body, luggage) of a randomly chosen passenger. Assume that $\mu_X = 72$ kg, $\sigma_X = 14$ kg, $\mu_Y = 12$ kg, $\sigma_Y = 3$ kg, and $\rho_{X,Y} = 0.3$.

- What is the probability that the cargo carrying capacity is exceeded if there are 120 random passengers?
- Determine the maximum number of passengers such that the cargo carrying capacity is exceeded with at most 1% probability.
- What is the (new) cargo carrying capacity if the probability in (a) equals 1%?

Tip: Mean and Standard Deviation of a Linear Combination

For any linear combination $W = aX + bY + c$ of two random variables X and Y , the mean and standard deviation of W are:

$$\mu_W = a\mu_X + b\mu_Y + c, \quad \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho_{X,Y}\sigma_X\sigma_Y}$$

This holds generally - whether or not X and Y are normal, correlated, or independent. The constant c never affects the spread, only the mean. If X and Y are independent, $\rho_{X,Y} = 0$ and the cross term drops out.

Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only

approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 20: (a) Probability the cargo capacity is exceeded with 120 passengers

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random passenger and record their body weight and luggage weight.

- **Mother variables:**

X	body weight of a random passenger (in kg)
Y	luggage weight of the same random passenger (in kg)

- **Other random variables:**

$P = X + Y$	total weight of a random passenger (in kg)
W_n	total weight of n random passengers (in kg)

• **Given:**

$$\mu_X = 72$$

population mean body weight of all passengers

$$\sigma_X = 14$$

population standard deviation of body weight of all passengers

$$\mu_Y = 12$$

population mean luggage weight of all passengers

$$\sigma_Y = 3$$

population standard deviation of luggage weight of all passengers

$$\rho_{X,Y} = 0.3$$

population correlation between all passengers' body and luggage weight

the cargo carrying capacity is 10 000 kg (10 tons)

all passengers' total weights are independent

.....

Step 1: Find the mean and standard deviation of a single passenger's total weight.

With $a = 1$, $b = 1$, $c = 0$:

$$\mu_P = 72 + 12 = 84$$

$$\sigma_P = \sqrt{14^2 + 3^2 + 2 \cdot 0.3 \cdot 14 \cdot 3} \approx 15.1723$$

Step 2: Apply the Central Limit Theorem to the total weight of 120 passengers.

A single passenger's total weight isn't necessarily normal, so we check the conditions for the Central Limit Theorem:

$$\left. \begin{array}{l} P_1, \dots, P_{120} \text{ non normal} \\ \text{independent} \\ n = 120 \geq 30 \\ E(P) = 84, \sigma_P = 15.1723 \end{array} \right\} \Rightarrow \begin{array}{l} W_{120} \approx \mathcal{N}(120 \cdot 84, \sqrt{120} \cdot 15.1723) \\ = \mathcal{N}(10080, 166.2047) \end{array}$$

Step 3: Compute the probability.

With W_{120} 's distribution in hand, we find the probability it exceeds the 10 000 kg capacity:

$$P(W_{120} > 10000) = \text{normalcdf}(10000, E99, 10080, 166.2047) \approx 0.6849$$

Answer: The probability that the cargo capacity is exceeded is 0.6849.

Tip: Finding the Largest n for a Probability Condition

When a question asks for the largest sample size n satisfying a probability condition (e.g. $P(\dots) \leq 0.01$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the largest X for which Y_1 still satisfies the condition.

Solution 20: (b) Maximum number of passengers so the capacity is exceeded with at most 1% probability

Step 1: Generalize the distribution of the total weight to an arbitrary n .

Repeating part (a)'s reasoning with n left symbolic:

$$W_n \approx \mathcal{N}(84n, \sqrt{230.2n})$$

Step 2: Search for the largest n .

We want the largest n such that:

$$P(W_n > 10000) \leq 0.01$$

We enter this into Y_1 :

$$Y_1 = \text{normalcdf}(10000, E99, 84X, \sqrt{230.2X})$$

Searching around $n = 114$ using the table, we find that $n = 114$ is the largest value for which the condition still holds:

$$n = 114 \Rightarrow P(W_{114} > 10000) \approx 0.0044 \leq 0.01$$

$$n = 115 \Rightarrow P(W_{115} > 10000) \approx 0.0183 > 0.01$$

Answer: The maximum number of passengers is 114.

✓ Solution 20: (c) New cargo capacity so the probability in (a) equals 1%

Step 1: Set up the equation.

We want $P(W_{120} > C) = 0.01$, with C unknown, where $W_{120} \sim \mathcal{N}(10080, 166.2047)$ as found in part (a):

$$\text{normalcdf}(X, E99, 10080, 166.2047) = 0.01$$

Step 2: Solve using the numeric solver.

Set E1: $\text{normalcdf}(X, E99, 10080, 166.2047)$, E2: 0.01, an initial guess $X = 10000$, then SOLVE:

$$C \approx 10466.65$$

Answer: The new cargo carrying capacity should be 10467 kg.

Question 21

The time (in hours) a random student requires to solve a statistics exam is measured by a continuous random variable X with the probability density function:

$$f_X(x) = \begin{cases} ax^3(5-x)^2 & \text{if } 0 \leq x \leq 5, \\ 0 & \text{otherwise.} \end{cases}$$

The maximum available time to solve the exam is 3 hours. What is the probability that the average required time in a random sample of 300 students exceeds 3 hours?

Tip: Finding the Constant in a Density

For $f(x)$ to be a valid density, the total area under it must equal 1:

$$\int_a^b f(x) dx = 1$$

Enter the formula (without the constant) into Y_1 , use $\text{fnInt}(Y_1, X, a, b)$ to get a number in terms of the constant, then solve for the constant so the total equals 1. We never integrate by hand - always fnInt on the calculator.

Tip: Mean of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **mean** is:

$$\mu = \int_a^b x \cdot f(x) dx = \text{fnInt}(X \cdot Y_1, X, a, b)$$

We never compute this integral by hand - we always use fnInt on the calculator.

💡 Tip: Variance of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **variance** is:

$$\sigma^2 = \int_a^b x^2 \cdot f(x) dx - \mu^2 = \text{fnInt}(X^2 \cdot Y_1, X, a, b) - \mu^2$$

Just like the mean, we never compute this integral by hand - we always use `fnInt` on the calculator.

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 21:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random student and record how long they take to solve the exam.

- **Mother variable:**

X time required by a random student to solve the exam (in hours)

- **Other random variables:**

$\bar{X}_{300}$ mean time required by a random sample of 300 students (in hours)

- **Given:**

$$f(x) = \begin{cases} ax^3(5-x)^2 & 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

Step 1: Find the constant a .

X is defined on $[0, 5]$, so we set the total area under f equal to 1, factor out the constant a , and solve using fnInt:

$$\int_0^5 a \cdot x^3(5-x)^2 dx = 1 \quad \Rightarrow \quad a \cdot \int_0^5 x^3(5-x)^2 dx = 1$$

$$a = \frac{1}{\int_0^5 x^3(5-x)^2 dx} = \frac{1}{\text{fnInt}(X^3(5-X)^2, X, 0, 5)} = \frac{12}{3125} = 0.00384$$

Step 2: Find the mean and standard deviation of a single student's time.

With a known, we enter the full density into Y_1 :

$$Y_1 = \frac{12}{3125} x^3(5-x)^2$$

Integrate $x \cdot f(x)$ over the domain $[0, 5]$ to get the mean:

$$\mu = \text{fnInt}(X \cdot Y_1, X, 0, 5) = \frac{20}{7} \approx 2.8571$$

Integrate $x^2 \cdot f(x)$ the same way, then subtract μ^2 to get the variance, and take the square root for the standard deviation:

$$E(X^2) = \text{fnInt}(X^2 \cdot Y_1, X, 0, 5) \approx 8.9286$$

$$\sigma^2 = 8.9286 - (2.8571)^2 \approx 0.7653$$

$$\sigma = \sqrt{0.7653} \approx 0.8748$$

Step 3: Find the distribution of the mean time.

A single student's time isn't normal, but we're averaging $n = 300$ of them, and $300 \geq 30$, so the Central Limit Theorem applies:

$$\left. \begin{array}{l} X_1, \dots, X_{300} \text{ non normal} \\ \text{independent} \\ n = 300 \geq 30 \\ E(X) = 2.8571, \sigma_X \approx 0.8748 \end{array} \right\} \Rightarrow \begin{array}{l} \bar{X}_{300} \approx \mathcal{N}\left(2.8571, \frac{0.8748}{\sqrt{300}}\right) \\ = \mathcal{N}(2.8571, 0.0505) \end{array}$$

Step 4: Compute the probability.

Now that we know the distribution of $\bar{X}_{300}$, we can find the probability that it exceeds 3 hours:

$$P(\bar{X}_{300} > 3) \approx \text{normalcdf}(3, \text{E99}, 2.8571, 0.0505) \approx 0.00234$$

Answer: The probability that the mean required time exceeds 3 hours is 0.00234.

? Question 22

The amount of memory (in megabytes, MB) occupied by a random digital photo is a continuous random variable X with the probability density function:

$$f_X(x) = \begin{cases} ax^5(5-x)^2 & \text{if } 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

A memory card has a capacity of 1 gigabyte (1 GB = 1024 MB).

- (a) What is the probability that the total size of 315 random digital photos exceeds the capacity of the memory card?
- (b) What is the maximum number of photos that can be put on the memory card such that the probability of not exceeding the capacity is at least 93%?

💡 Tip: Finding the Constant in a Density

For $f(x)$ to be a valid density, the total area under it must equal 1:

$$\int_a^b f(x) dx = 1$$

Enter the formula (without the constant) into Y_1 , use $\text{fnInt}(Y_1, X, a, b)$ to get a number in terms of the constant, then solve for the constant so the total equals 1. We never integrate by hand - always fnInt on the calculator.

💡 Tip: Mean of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **mean** is:

$$\mu = \int_a^b x \cdot f(x) dx = \text{fnInt}(X \cdot Y_1, X, a, b)$$

We never compute this integral by hand - we always use `fnInt` on the calculator.

💡 Tip: Variance of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **variance** is:

$$\sigma^2 = \int_a^b x^2 \cdot f(x) dx - \mu^2 = \text{fnInt}(X^2 \cdot Y_1, X, a, b) - \mu^2$$

Just like the mean, we never compute this integral by hand - we always use `fnInt` on the calculator.

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n} \sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies

in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 22: (a) Probability that 315 photos exceed the memory card's 1024 MB capacity

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random digital photo and record how much memory it uses.
- **Mother variable:**

X memory used by a random photo (in MB)

- **Other random variables:**

W_{315} total memory used by 315 random photos (in MB)

- **Given:**

$$f(x) = \begin{cases} ax^5(5-x)^2 & 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

the memory card's capacity is 1024 MB

photo sizes are independent

Step 1: Find the constant a .

X is defined on $[0, 5]$, so we set the total area under f equal to 1, factor out the constant a , and solve using fnInt:

$$\int_0^5 a \cdot x^5(5-x)^2 dx = 1 \quad \Rightarrow \quad a \cdot \int_0^5 x^5(5-x)^2 dx = 1$$

$$a = \frac{1}{\int_0^5 x^5(5-x)^2 dx} = \frac{1}{\text{fnInt}(X^5(5-X)^2, X, 0, 5)} \approx 0.00043008$$

Step 2: Find the mean and standard deviation of a single photo's memory usage.

With a known, we enter the full density into Y_1 , then integrate $x \cdot f(x)$ over the domain $[0, 5]$ to get the mean:

$$Y_1 = 0.00043008 x^5 (5 - x)^2$$

$$\mu = \text{fnInt}(X \cdot Y_1, X, 0, 5) \approx 3.3333$$

Integrate $x^2 \cdot f(x)$ the same way, then subtract μ^2 to get the variance, and take the square root for the standard deviation:

$$E(X^2) = \text{fnInt}(X^2 \cdot Y_1, X, 0, 5) \approx 11.6667$$

$$\sigma^2 = 11.6667 - (3.3333)^2 \approx 0.5556$$

$$\sigma = \sqrt{0.5556} \approx 0.7454$$

Step 3: Find the distribution of the total memory usage for 315 photos.

A single photo's memory usage isn't normal, but we're summing $n = 315$ of them, and $315 \geq 30$, so the Central Limit Theorem applies:

$$\left. \begin{array}{l} X_1, \dots, X_{315} \text{ non normal} \\ \text{independent} \\ n = 315 \geq 30 \\ E(X) = 3.3333, \sigma_X \approx 0.7454 \end{array} \right\} \Rightarrow \begin{array}{l} W_{315} \approx \mathcal{N}(315 \cdot 3.3333, \sqrt{315} \cdot 0.7454) \\ = \mathcal{N}(1050.00, 13.2312) \end{array}$$

Step 4: Compute the probability.

Now that we know the distribution of W_{315} , we can find the probability that it exceeds 1024 MB:

$$P(W_{315} > 1024) \approx \text{normalcdf}(1024, E99, 1050.00, 13.2312) \approx 0.9753$$

Answer: The probability of exceeding the memory card capacity is 0.9753.

💡 Tip: Finding the Largest n for a Probability Condition

When a question asks for the largest sample size n satisfying a probability condition (e.g. $P(\dots) \geq 0.93$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the largest X for which Y_1 still satisfies the condition.

✅ Solution 22: (b) Maximum number of photos so the capacity isn't exceeded with at least 93% probability

Step 1: Generalize the distribution of the total to an arbitrary n .

Repeating Step 3 from part (a), but keeping n symbolic instead of plugging in 315:

$$W_n \approx \mathcal{N}(3.3333n, 0.7454\sqrt{n})$$

Step 2: Search for the largest n .

We want the largest n such that:

$$P(W_n \leq 1024) \geq 0.93$$

We enter this into Y_1 :

$$Y_1 = \text{normalcdf}(-E99, 1024, 3.3333X, 0.7454\sqrt{X})$$

Searching around $n = 300$ using the table, we find that $n = 301$ is the largest value for which the condition still holds:

$$n = 301 \quad \Rightarrow \quad P(W_{301} \leq 1024) \approx 0.9450 \geq 0.93$$

Answer: A maximum of 301 photos can be stored with at least 93% probability.

Question 23

The number of winners in rank 1 of a single national lottery draw is a discrete random variable, X , with the following probability mass function:

k	0	1	2	3	4
$P(X = k)$	0.05	0.55	0.25	0.10	0.05

Assume that the numbers of winners in different draws are independent.

- What is the probability that in 1 week (consisting of 2 draws), there are at least 7 winners in total for rank 1?
- Calculate the probability that in 1 year (consisting of 104 draws), there are more than 150 winners in total for rank 1.
- How many draws are needed such that there is at least a 99% probability of having at least 100 winners in total for rank 1?

Tip: Mean of a Discrete Random Variable

Multiply each outcome by its own probability, then add:

$$\mu = E(X) = \sum_i x_i P(X=x_i)$$

Tip: Variance of a Discrete Random Variable - Convenience Formula

Do the same with each outcome squared to get $E(X^2)$, then subtract μ^2 :

$$\sigma^2 = E(X^2) - \mu^2, \quad \text{where } E(X^2) = \sum_i x_i^2 P(X=x_i)$$

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n} \sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 23: (a) Probability of at least 7 winners in 1 week (2 draws)

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random lottery draw and record the number of rank-1 winners.
- **Mother variable:**

X number of rank-1 winners in a random draw

- **Other random variables:**

W_n total number of rank-1 winners in n random draws

- **Given:** The probability distribution of X :

k	0	1	2	3	4
$P(X = k)$	0.05	0.55	0.25	0.10	0.05

The numbers of winners in different draws are independent.

.....

With only $n = 2$ draws, the sample size is too small for the Central Limit Theorem, so we compute the exact probability by considering every combination of the two draws (X_1, X_2) that totals 7 or more: $(3, 4)$, $(4, 3)$, and $(4, 4)$.

Since the draws are independent, we multiply their probabilities and add:

$$\begin{aligned}
 P(W_2 \geq 7) &= P(X_1=3, X_2=4) + P(X_1=4, X_2=3) + P(X_1=4, X_2=4) \\
 &= (0.10 \cdot 0.05) + (0.05 \cdot 0.10) + (0.05 \cdot 0.05) \\
 &= 0.005 + 0.005 + 0.0025 = 0.0125
 \end{aligned}$$

Answer: The probability of at least 7 winners in 2 draws is 0.0125.

✓ Solution 23: (b) Probability of more than 150 winners in 1 year (104 draws)

Step 1: Find the mean and standard deviation of a single draw's winners.

Multiply each outcome by its probability, then add:

$$\mu = E(X) = 0 \cdot 0.05 + 1 \cdot 0.55 + 2 \cdot 0.25 + 3 \cdot 0.10 + 4 \cdot 0.05 = 1.55$$

Then use the convenience formula for the variance:

$$\begin{aligned}
 \sigma^2 &= E(X^2) - \mu^2 \\
 &= \left(0^2 \cdot 0.05 + 1^2 \cdot 0.55 + 2^2 \cdot 0.25 + 3^2 \cdot 0.10 + 4^2 \cdot 0.05\right) - 1.55^2 \\
 &= 3.25 - 2.4025 = 0.8475 \\
 \sigma &= \sqrt{0.8475} \approx 0.9206
 \end{aligned}$$

Step 2: Apply the Central Limit Theorem to the total winners over 104 draws.

The number of winners in a draw is not necessarily normal, so we check the conditions for the Central Limit Theorem:

$$\left. \begin{array}{l} X_1, \dots, X_{104} \text{ non normal} \\ \text{independent} \\ n = 104 \geq 30 \\ E(X) = 1.55, \sigma_X \approx 0.9206 \end{array} \right\} \Rightarrow \begin{aligned} W_{104} &\approx \mathcal{N}(104 \cdot 1.55, \sqrt{104} \cdot 0.9206) \\ &= \mathcal{N}(161.2, 9.3883) \end{aligned}$$

Step 3: Compute the probability.

With W_{104} 's distribution in hand, we find the probability it exceeds 150:

$$P(W_{104} > 150) \approx \text{normalcdf}(150, \text{E99}, 161.2, 9.3883) \approx 0.8836$$

Answer: The probability of more than 150 winners is 0.8836.

💡 Tip: Finding the Smallest n for a Probability Condition

When a question asks for the smallest sample size n satisfying a probability condition (e.g. $P(\dots) \geq 0.99$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the smallest X for which Y_1 satisfies the condition.

✓ Solution 23: (c) Number of draws needed for at least 99% probability of at least 100 winners

Step 1: Generalize the distribution of the total winners to an arbitrary n .

Repeating part (b)'s reasoning with n left symbolic:

$$W_n \approx \mathcal{N}(1.55n, \sqrt{0.8475n})$$

Step 2: Search for the smallest n .

We want the smallest n such that:

$$P(W_n \geq 100) \geq 0.99$$

We enter this into Y_1 :

$$Y_1 = \text{normalcdf}(100, E99, 1.55X, \sqrt{0.8475X})$$

Searching around $n = 70$ using the table, we find that $n = 77$ is the smallest value for which the condition holds:

$$n = 77 \Rightarrow P(W_{77} \geq 100) \approx 0.9917 \geq 0.99$$

Answer: At least 77 draws are needed.

Question 24

Santa Claus orders crates, each filled with 30 mandarins. The weights of the mandarins and the empty crates are all independent and normally distributed. The weight of a single mandarin is a random variable, M , with an unknown population mean μ_M and a population standard deviation of $\sigma_M = 5$ grams. The weight of an empty crate is a random variable, C , with a population mean of $\mu_C = 50$ grams and a population standard deviation of $\sigma_C = 2$ grams.

- What is the probability that a random sample of 100 empty crates weighs more than 4950 grams in total?
- A supplier claims that 99% of their filled crates (an empty crate plus 30 mandarins) weigh at least 2.5 kilograms (1 kg = 1000 g). Calculate the minimal value for the mean weight of a single mandarin, μ_M , that would be consistent with this claim.

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

✓ Solution 24: (a) Probability that 100 empty crates weigh more than 4950 grams

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random empty crate and record its weight.

- **Mother variable:**

C weight of a random empty crate (in g)

- **Other random variables:**

W_{100} total weight of 100 random empty crates (in g)

- **Given:**

$\mu_C = 50$ population mean weight of all empty crates

$\sigma_C = 2$ population standard deviation of weight of all empty crates

$C \sim \mathcal{N}(\mu_C = 50, \sigma_C = 2)$

crate weights are independent

.....

Step 1: Find the distribution of the total weight of 100 empty crates.

The weight of one empty crate is normal. We're looking for the sum of 100 crates, so:

$$\left. \begin{array}{l} C_1, \dots, C_{100} \sim \mathcal{N}(50, 2) \\ \text{independent} \end{array} \right\} \Rightarrow W_{100} \sim \mathcal{N}(100 \cdot 50, \sqrt{100} \cdot 2) = \mathcal{N}(5000, 20)$$

Step 2: Compute the probability.

With W_{100} 's distribution in hand, we find the probability it's more than 4950:

$$P(W_{100} > 4950) = \text{normalcdf}(4950, \text{E99}, 5000, 20) \approx 0.9938$$

Answer: The probability that 100 empty crates weigh more than 4950 grams is 0.9938.

💡 Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ Solution 24: (b) Minimum mean mandarin weight consistent with the supplier's claim

For this part we also need the mandarin weight:

M weight of a random mandarin (in g)

$$M \sim \mathcal{N}(\mu_M = ?, \sigma_M = 5)$$

W_M total weight of the 30 mandarins in a crate (in g)

$T = C + W_M$ total weight of a filled crate (in g)

A filled crate contains 1 empty crate and 30 mandarins, all independent.

Step 1: Find the distribution of the total mandarin weight in one crate.

The weight of one mandarin is normal. We're looking for the sum of 30 mandarins, so:

$$\left. \begin{array}{l} M_1, \dots, M_{30} \sim \mathcal{N}(\mu_M, 5) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_M \sim \mathcal{N}(30\mu_M, \sqrt{30} \cdot 5) \\ = \mathcal{N}(30\mu_M, 27.3861) \end{array}$$

Step 2: Find the distribution of the total filled crate weight.

This is a linear combination of C and W_M , with $a = 1, b = 1, c = 0$, and $\rho = 0$ since they're

independent:

$$\mu_T = 50 + 30\mu_M$$

$$\sigma_T = \sqrt{2^2 + 27.3861^2} \approx 27.4591$$

$$T \sim \mathcal{N}(50 + 30\mu_M, 27.4591)$$

Step 3: Find μ_M using the numeric solver.

We want the smallest μ_M such that:

$$P(T \geq 2500) = 0.99$$

Enter the equation directly into the numeric solver, solving for the unknown mean, X:

$$\text{normalcdf}(2500, \text{E99}, 30X + 50, 27.4591) = 0.99$$

Set E1: normalcdf(2500, E99, 30X+50, 27.4591), E2: 0.99, an initial guess X = 50, then SOLVE:

$$\mu_M \approx 83.80$$

Answer: The minimum mean weight of a single mandarin must be 83.80 grams.

? Question 25

The lengths of man's trousers are normally distributed with mean 90 cm and standard deviation 8 cm. The lengths of woman's trousers are normally distributed with mean 75 cm and standard deviation 5 cm.

- (a) Calculate the probability that a random woman's trousers is longer than a random man's trousers.
- (b) What is the probability that 50 man's trousers and 40 woman's trousers have a total length of at least 77 metres (1 m = 100 cm)?
- (c) A youth movement wants to build a chain of trousers with the same number of man's trousers and woman's trousers. How many trousers do they need such that this chain is at least 1 kilometre (1 km = 100,000 cm) long with at least 99% probability?

💡 Tip: Comparing Two Variables

To find a probability comparing two variables (e.g. $P(X > Y)$), move everything to one side first, turning it into a statement about their difference:

$$P(X > Y) = P(X - Y > 0)$$

Then find the distribution of $X - Y$ using the linear combination formula for two normal variables.

💡 Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✅ Solution 25: (a) Probability a random woman's trousers is longer than a random man's

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random man's trousers and a random woman's trousers and record their lengths.
- **Mother variables:**

X	length of a random man's trousers (in cm)
Y	length of a random woman's trousers (in cm)

• **Given:**

$$\mu_X = 90$$

population mean length of all men's trousers

$$\sigma_X = 8$$

population standard deviation of length of all men's trousers

$$\mu_Y = 75$$

population mean length of all women's trousers

$$\sigma_Y = 5$$

population standard deviation of length of all women's trousers

$$X \sim \mathcal{N}(\mu_X = 90, \sigma_X = 8)$$

$$Y \sim \mathcal{N}(\mu_Y = 75, \sigma_Y = 5)$$

all trouser lengths are independent

.....
Step 1: Rewrite the comparison as a statement about the difference.

We want a woman's trousers to be longer than a man's, so we move everything to one side:

$$P(Y > X) = P(Y - X > 0)$$

Step 2: Find the distribution of the difference.

This is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho_{X,Y} = 0$ since X and Y are independent:

$$\mu_{Y-X} = 75 - 90 = -15$$

$$\sigma_{Y-X} = \sqrt{5^2 + 8^2} = \sqrt{89} \approx 9.4340$$

So $Y - X \sim \mathcal{N}(-15, 9.4340)$.

Step 3: Compute the probability.

With the distribution of $Y - X$ in hand:

$$P(Y - X > 0) = \text{normalcdf}(0, \infty, -15, 9.4340) \approx 0.0559$$

Answer: The probability is 0.0559.

💡 Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left. \begin{array}{l} X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma) \\ \text{independent} \end{array} \right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

✅ Solution 25: (b) Probability that 50 men's and 40 women's trousers total at least 77 metres

Step 1: Find the distribution of the total length of 50 men's trousers.

The length of one man's trousers is normal. We're looking for the sum of 50 of them, so:

$$\left. \begin{array}{l} X_1, \dots, X_{50} \sim \mathcal{N}(90, 8) \\ \text{independent} \end{array} \right\} \Rightarrow W_X \sim \mathcal{N}(50 \cdot 90, \sqrt{50} \cdot 8) = \mathcal{N}(4500, 56.5685)$$

Step 2: Find the distribution of the total length of 40 women's trousers.

The length of one woman's trousers is normal too. We're looking for the sum of 40 of them, so:

$$\left. \begin{array}{l} Y_1, \dots, Y_{40} \sim \mathcal{N}(75, 5) \\ \text{independent} \end{array} \right\} \Rightarrow W_Y \sim \mathcal{N}(40 \cdot 75, \sqrt{40} \cdot 5) = \mathcal{N}(3000, 31.6228)$$

Step 3: Find the distribution of the combined total length.

The combined total length is $T = W_X + W_Y$. This is a linear combination with $a = 1, b = 1$,

$c = 0$, and $\rho = 0$ since they're independent:

$$\mu_T = 4500 + 3000 = 7500$$

$$\sigma_T = \sqrt{56.5685^2 + 31.6228^2} \approx 64.8074$$

So $T \sim \mathcal{N}(7500, 64.8074)$.

Step 4: Compute the probability.

We need the total length to be at least 77 metres, i.e. 7700 cm:

$$P(T \geq 7700) = \text{normalcdf}(7700, E99, 7500, 64.8074) \approx 0.001014$$

Answer: The probability is 0.001014.

 Tip: Finding the Smallest n for a Probability Condition

When a question asks for the smallest sample size n satisfying a probability condition (e.g. $P(\dots) \geq 0.99$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the smallest X for which Y_1 satisfies the condition.

✓ **Solution 25: (c) Minimum number of trousers from each group for a 1 km chain, with 99% probability**

Step 1: Generalize the distribution of the combined total to an arbitrary n .

Repeating Steps 1-2 from part (b), but keeping n symbolic instead of plugging in 50 and 40 (both groups now use the same size n):

$$W_X \sim \mathcal{N}(90n, 8\sqrt{n}), \quad W_Y \sim \mathcal{N}(75n, 5\sqrt{n})$$

The combined total is $W_X + W_Y$. This is a linear combination with $a = 1, b = 1, c = 0$:

$$\mu_{W_X+W_Y} = 90n + 75n = 165n$$

$$\sigma_{W_X+W_Y} = \sqrt{(8\sqrt{n})^2 + (5\sqrt{n})^2} = \sqrt{89n}$$

So $W_X + W_Y \sim \mathcal{N}(165n, \sqrt{89n})$.

Step 2: Search for the smallest n .

We want the smallest n such that the chain (1 km = 100 000 cm) is reached with at least 99% probability:

$$P(W_X + W_Y \geq 100000) \geq 0.99$$

We enter this into Y_1 :

$$Y_1 = \text{normalcdf}(100000, E99, 165X, \sqrt{89X})$$

Searching around $n = 600$ using the table, we find that $n = 610$ is the smallest value for which the condition holds:

$$n = 610 \Rightarrow P(W_X + W_Y \geq 100000) \approx 0.9974 \geq 0.99$$

Since the chain needs n trousers from *each* group, the total number of trousers is $2n$:

$$2 \cdot 610 = 1220$$

Answer: At least 1220 trousers (610 men's + 610 women's) are needed.

Question 26

The number of bicycles stolen from a randomly chosen KU Leuven (KUL) student during their student career is a discrete random variable, X , with the following probability mass function:

k	0	1	2	3	4
$P(X = k)$	0.35	0.25	0.20	0.15	0.05

Assume that the numbers of stolen bicycles for different students are independent.

- What is the probability that a total of less than 120 bicycles will be stolen from a random sample of 100 KUL students?
- Consider two independent random samples, each of 100 KUL students. What is the probability that the total number of stolen bicycles in the two samples differs by more than 10?
- What is the probability that, in a random sample of 100 KUL students, more than one-third of them have had more than 1 bicycle stolen?

Tip: Mean of a Discrete Random Variable

Multiply each outcome by its own probability, then add:

$$\mu = E(X) = \sum_i x_i P(X=x_i)$$

Tip: Variance of a Discrete Random Variable - Convenience Formula

Do the same with each outcome squared to get $E(X^2)$, then subtract μ^2 :

$$\sigma^2 = E(X^2) - \mu^2, \quad \text{where } E(X^2) = \sum_i x_i^2 P(X=x_i)$$

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 26: (a) Probability that fewer than 120 bicycles are stolen from 100 students

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random KUL student and record how many bicycles were stolen from them.
- **Mother variable:**

X number of bicycles stolen from a random student

- **Other random variables:**

W_{100} total number of bicycles stolen from 100 random students

- **Given:** The probability distribution of X :

k	0	1	2	3	4
$P(X = k)$	0.35	0.25	0.20	0.15	0.05

The numbers of stolen bicycles for different students are independent.

Step 1: Find the mean and standard deviation of the number of bicycles stolen from a single student.

Multiply each outcome by its probability, then add:

$$\mu = E(X) = 0 \cdot 0.35 + 1 \cdot 0.25 + 2 \cdot 0.20 + 3 \cdot 0.15 + 4 \cdot 0.05 = 1.3$$

Then use the convenience formula for the variance:

$$\begin{aligned}\sigma^2 &= E(X^2) - \mu^2 \\ &= (0 \cdot 0.35 + 1 \cdot 0.25 + 4 \cdot 0.20 + 9 \cdot 0.15 + 16 \cdot 0.05) - 1.3^2 \\ &= 3.2 - 1.69 = 1.51 \\ \sigma &= \sqrt{1.51} \approx 1.2288\end{aligned}$$

Step 2: Find the distribution of the total number of stolen bicycles.

The number of bicycles stolen from a single student isn't normal, but we're summing $n = 100$ students, and $100 \geq 30$, so the Central Limit Theorem applies:

$$\left. \begin{array}{l} X_1, \dots, X_{100} \text{ non normal} \\ \text{independent} \\ n = 100 \geq 30 \\ E(X) = 1.3, \sigma_X \approx 1.2288 \end{array} \right\} \Rightarrow \begin{aligned} W_{100} &\approx \mathcal{N}(100 \cdot 1.3, \sqrt{100} \cdot 1.2288) \\ &= \mathcal{N}(130, 12.288) \end{aligned}$$

Step 3: Compute the probability.

With W_{100} 's distribution in hand, we find the probability it's below 120:

$$P(W_{100} < 120) \approx \text{normalcdf}(-E99, 120, 130, 12.288) \approx 0.2079$$

Answer: The probability is 0.2079.

💡 Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ Solution 26: (b) Probability the two samples' totals deviate by more than 10 bikes

Step 1: Translate the event into two one-sided pieces.

The difference being "more than 10" doesn't say which sample has the bigger total - either the first could exceed the second by more than 10, or the second could exceed the first by more than 10. These are two separate, non-overlapping events, so we add their probabilities instead of using absolute value notation:

$$P(W_{100} - W'_{100} > 10) + P(W_{100} - W'_{100} < -10)$$

Step 2: Find the distribution of the first sample's total.

This is the same computation as part (a):

$$W_{100} \approx \mathcal{N}(130, 12.288)$$

Step 3: Find the distribution of the second sample's total.

The second sample of 100 students is chosen the exact same way, independently of the first:

$$W'_{100} \approx \mathcal{N}(130, 12.288)$$

Step 4: Find the distribution of the difference.

This is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho = 0$ since the two samples are independent:

$$\mu_{W_{100} - W'_{100}} = 130 - 130 = 0$$

$$\sigma_{W_{100} - W'_{100}} = \sqrt{12.288^2 + 12.288^2} \approx 17.3785$$

So $W_{100} - W'_{100} \approx \mathcal{N}(0, 17.3785)$.

Step 5: Compute the probability.

With the distribution of $W_{100} - W'_{100}$ in hand:

$$P(W_{100} - W'_{100} > 10) \approx \text{normalcdf}(10, \text{E99}, 0, 17.3785) \approx 0.2825$$

$$P(W_{100} - W'_{100} < -10) \approx \text{normalcdf}(-\text{E99}, -10, 0, 17.3785) \approx 0.2825$$

This makes sense by symmetry: since the difference is centered at 0, both tails beyond ± 10 carry exactly the same probability.

Now we find the total:

$$P(W_{100} - W'_{100} > 10) + P(W_{100} - W'_{100} < -10) \approx 0.2825 + 0.2825 = 0.5650$$

Answer: The probability is 0.5650.

✓ Solution 26: (c) Probability more than a third of 100 students had more than 1 bike stolen

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random KUL student and record whether they had more than 1 bicycle stolen.
- **Mother variable:**

X number of bicycles stolen from a random student

- **Other random variables:**

Y number of students (out of 100) who had more than 1 bicycle stolen

- **Given:** the distribution of X , same as in part (a).

.....
Notice that Y is a more general version of X : each of the 100 students either did or didn't have more than 1 bicycle stolen, and we're counting how many did. This sounds like a binomial setup - but first, we need the probability π for a single student.

Step 1: Find the single-trial probability π .

From the table:

$$\pi = P(X > 1) = 0.20 + 0.15 + 0.05 = 0.40$$

Step 2: Plug π into the binomial formula.

Check the 3 conditions for a binomial variable:

- The underlying yes/no event is a student having more than 1 bicycle stolen, with $\pi = 0.40$.
- We check for 100 students (one per student sampled).
- The students are independent of each other.

All 3 hold, so $Y \sim \text{Bin}(100, 0.40)$. "More than a third" of 100 students means more than $33.\bar{3}$, so since Y only takes whole values, this means $Y \geq 34$:

$$P(Y \geq 34) = 1 - \text{binomcdf}(100, 0.40, 33) \approx 0.9087$$

Answer: The probability is 0.9087.

? Question 27

Solve the previous exercise with $X \sim \mathcal{B}(4, 0.325)$.

💡 Tip: Mean and Standard Deviation of a Binomial Variable

If $X \sim \text{Bin}(n, \pi)$, its mean and standard deviation are:

$$\mu = n\pi, \quad \sigma = \sqrt{n\pi(1 - \pi)}$$

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ Solution 27: (a) Probability that fewer than 120 bicycles are stolen from 100 students

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random KUL student and record how many bicycles were stolen from them.
- **Mother variable:**

X number of bicycles stolen from a random student

- **Other random variables:**

W_{100} total number of bicycles stolen from 100 random students

- **Given:**

$n = 4$ number of Bernoulli trials underlying X 's binomial distribution

$\pi = 0.325$ single-trial success probability underlying X 's binomial distribution

$X \sim \text{Bin}(4, 0.325)$
different students are independent

.....

Step 1: Find the mean and standard deviation of a single student's stolen bikes.

Using the binomial mean and standard deviation formulas:

$$\mu = 4 \cdot 0.325 = 1.3, \quad \sigma = \sqrt{4 \cdot 0.325 \cdot 0.675} \approx 0.9367$$

Step 2: Notice an exact method is available here.

Unlike the general discrete distribution in the previous exercise, $X \sim \text{Bin}(4, 0.325)$ is itself Binomial. The sum of 100 independent copies of a Binomial(4, 0.325) variable is itself exactly

Binomial - specifically, $W_{100} \sim \text{Bin}(400, 0.325)$, since it counts the total successes across $100 \times 4 = 400$ independent Bernoulli trials, each with success probability 0.325.

Method 1: Exact, using the Binomial Distribution.

We compute this directly with the calculator, no approximation needed:

$$P(W_{100} < 120) = P(W_{100} \leq 119) = \text{binomcdf}(400, 0.325, 119) \approx 0.1308$$

Method 2: Approximate, using the Central Limit Theorem.

The number of bicycles stolen from a single student isn't normal, but we're summing $n = 100$ students, and $100 \geq 30$, so the Central Limit Theorem also applies:

$$\left. \begin{array}{l} X_1, \dots, X_{100} \text{ non normal} \\ \text{independent} \\ n = 100 \geq 30 \\ E(X) = 1.3, \sigma_X \approx 0.9367 \end{array} \right\} \Rightarrow \begin{array}{l} W_{100} \approx \mathcal{N}(100 \cdot 1.3, \sqrt{100} \cdot 0.9367) \\ = \mathcal{N}(130, 9.3675) \end{array}$$

With W_{100} 's distribution in hand, we find the probability it's below 120:

$$P(W_{100} < 120) \approx \text{normalcdf}(-E99, 120, 130, 9.3675) \approx 0.1429$$

Answer: The exact probability is 0.1308. The approximate probability is 0.1429.

💡 Tip: Linear Combination of Two Normal Variables

For any constants a, b, c , and $W = aX + bY + c$:

$$\left. \begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array} \right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_X\sigma_Y\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

✓ Solution 27: (b) Probability the two samples' totals deviate by more than 10 bikes

Step 1: Translate the event into two one-sided pieces.

The difference being "more than 10" doesn't say which sample has the bigger total - either the first could exceed the second by more than 10, or the second could exceed the first by more than 10. These are two separate, non-overlapping events, so we add their probabilities instead of using absolute value notation:

$$P(W_{100} - W'_{100} > 10) + P(W_{100} - W'_{100} < -10)$$

Step 2: Find the distribution of the first sample's total.

This is the same computation as part (a):

$$W_{100} \approx \mathcal{N}(130, 9.3675)$$

Step 3: Find the distribution of the second sample's total.

The second sample of 100 students is chosen the exact same way, independently of the first:

$$W'_{100} \approx \mathcal{N}(130, 9.3675)$$

Step 4: Find the distribution of the difference.

This is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho = 0$ since the two samples are independent:

$$\mu_{W_{100} - W'_{100}} = 130 - 130 = 0$$

$$\sigma_{W_{100} - W'_{100}} = \sqrt{9.3675^2 + 9.3675^2} \approx 13.2476$$

So $W_{100} - W'_{100} \approx \mathcal{N}(0, 13.2476)$.

Step 5: Compute the probability.

With the distribution of $W_{100} - W'_{100}$ in hand:

$$P(W_{100} - W'_{100} > 10) \approx \text{normalcdf}(10, \text{E99}, 0, 13.2476) \approx 0.2252$$

$$P(W_{100} - W'_{100} < -10) \approx \text{normalcdf}(-\text{E99}, -10, 0, 13.2476) \approx 0.2252$$

This makes sense by symmetry: since the difference is centered at 0, both tails beyond ± 10 carry exactly the same probability.

Now we find the total:

$$P(W_{100} - W'_{100} > 10) + P(W_{100} - W'_{100} < -10) \approx 0.2252 + 0.2252 = 0.4503$$

Answer: The probability is 0.4503.

✓ Solution 27: (c) Probability more than a third of 100 students had more than 1 bike stolen

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random KUL student and record whether they had more than 1 bicycle stolen.
- **Mother variable:**

X number of bicycles stolen from a random student

- **Other random variables:**

Y number of students (out of 100) who had more than 1 bicycle stolen

- **Given:** $X \sim \text{Bin}(4, 0.325)$, same as in part (a).

.....

Notice that Y is a more general version of X : each of the 100 students either did or didn't have more than 1 bicycle stolen, and we're counting how many did. This sounds like a binomial setup - but first, we need the probability π for a single student.

Step 1: Find the single-trial probability π .

Using the given distribution of X :

$$\pi = P(X > 1) = 1 - P(X \leq 1) = 1 - \text{binomcdf}(4, 0.325, 1) \approx 0.3926$$

Step 2: Plug π into the binomial formula.

Check the 3 conditions for a binomial variable:

- The underlying yes/no event is a student having more than 1 bicycle stolen, with $\pi \approx 0.3926$.
- We check for 100 students (one per student sampled).
- The students are independent of each other.

All 3 hold, so $Y \sim \text{Bin}(100, 0.3926)$. "More than a third" of 100 students means more than $33.\bar{3}$, so since Y only takes whole values, this means $Y \geq 34$:

$$P(Y \geq 34) = 1 - \text{binomcdf}(100, 0.3926, 33) \approx 0.8816$$

Answer: The probability is 0.8816.

Question 28

In the final of the snooker world championship, the contestants have to play a match according to the best of 35 frames rule, i.e., the player who first wins 18 frames is the winner of the match. Suppose that Ronnie O'Sullivan and John Higgins play the final and that Ronnie wins each frame with probability 0.6, independent of the other frames.

What is the probability that Ronnie O'Sullivan wins the match? Give both an exact and an approximate solution.

Tip: Binomial Distribution

Use the binomial distribution when a trial is repeated a fixed number of times, independently, with the same success probability each time:

$$Y \sim \text{Bin}(n, \pi), \quad P(Y = k) = \text{binompdf}(n, \pi, k)$$

Tip: Normal Approximation to the Binomial

If $Y \sim \text{Bin}(n, \pi)$, we can approximate it using a normal distribution with matching mean and standard deviation:

$$Y \approx \mathcal{N}\left(n\pi, \sqrt{n\pi(1-\pi)}\right)$$

This approximation is only valid when **both** of the following hold:

- $n\pi \geq 10$
- $n(1 - \pi) \geq 10$

✓ Solution 28:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random frame of the match and record whether Ronnie wins it.

- **Mother variable:**

$$U = \begin{cases} 1 & \text{if Ronnie wins a random frame} \\ 0 & \text{if Ronnie doesn't win a random frame} \end{cases}$$

- **Other random variables:**

$$X \quad \text{number of frames (out of 35) Ronnie wins}$$

- **Given:**

$$\pi = 0.6 \quad \text{probability Ronnie wins a random frame}$$

frame outcomes are independent

.....

For Ronnie to win the match, he must win at least 18 of the 35 possible frames: even though the real match stops early once someone reaches 18, imagining all 35 frames played out, Ronnie winning 18 or more of them is exactly equivalent to him winning the match. So we want $P(X \geq 18)$.

Step 1: Check the conditions for a binomial variable.

We check the 3 conditions:

- The underlying yes/no event is Ronnie winning a frame, with $\pi = 0.6$.
- It happens $n = 35$ times (one per frame).
- The frames are independent of each other.

All 3 hold, so $X \sim \text{Bin}(35, 0.6)$.

Method 1: Exact, using the Binomial Distribution.

We compute this directly with the calculator, no approximation needed:

$$P(X \geq 18) = 1 - P(X \leq 17) = 1 - \text{binomcdf}(35, 0.6, 17) \approx 0.8857$$

Method 2: Approximate, using the normal Distribution.

Check both conditions for the normal approximation:

$$n\pi = 35 \cdot 0.6 = 21 \geq 10, \quad n(1 - \pi) = 35 \cdot 0.4 = 14 \geq 10$$

Both hold, so:

$$X \approx \mathcal{N}(n\pi, \sqrt{n\pi(1 - \pi)}) = \mathcal{N}(21, \sqrt{8.4})$$

We find the probability that it's at least 18:

$$P(X \geq 18) \approx \text{normalcdf}(18, \text{E99}, 21, \sqrt{8.4}) \approx 0.8497$$

Answer: The exact probability is 0.8857. The approximate probability is 0.8497.

Question 29

The operational times of light bulbs are exponentially distributed with an expectation of 5000 hours. How many random light bulbs do I need to buy such that I am at least 90% sure that I can light my basement without interruption during a period of 20 years?

Hint: The expectation and the standard deviation are equal in the case of an exponential distribution.

Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

💡 Tip: Finding the Smallest n for a Probability Condition

When a question asks for the smallest sample size n satisfying a probability condition (e.g. $P(\dots) \geq 0.90$), use the calculator's table to test values of n directly:

- Press $Y=$ and enter the condition's probability into Y_1 , with X in place of n .
- Press $2ND \rightarrow TBLSET$, and set $TblStart$ to a reasonable starting guess for n .
- Press $2ND \rightarrow TABLE$, and scroll until you find the smallest X for which Y_1 satisfies the condition.

✓ Solution 29:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random light bulb and record its operational time.
- **Mother variable:**

X operational time of a random light bulb (in hours)

- **Other random variables:**

W_n total operational time of n random light bulbs (in hours)

- **Given:**

$\mu = 5000$ population mean lifetime of all bulbs

$\sigma = 5000$ population standard deviation of lifetime of all bulbs

the required duration is 175 200 hours ($20 \cdot 365 \cdot 24$)

X is exponentially distributed

bulb lifetimes are independent

Step 1: Generalize the distribution of the total lifetime to an arbitrary n .

A single bulb's lifetime is exponential (skewed, not necessarily normal), but we expect to need a fairly large number of bulbs, so the Central Limit Theorem applies:

$$W_n \approx \mathcal{N}(5000n, 5000\sqrt{n})$$

Step 2: Search for the smallest n .

We want the smallest n such that:

$$P(W_n \geq 175200) \geq 0.90$$

We enter this into Y₁:

$$Y_1 = \text{normalcdf}(175200, E99, 5000X, 5000\sqrt{X})$$

Searching around $n = 40$ using the table, we find that $n = 44$ is the smallest value for which the condition holds:

$$n = 44 \Rightarrow P(W_{44} \geq 175200) \approx 0.9116 \geq 0.90$$

Answer: You need at least 44 light bulbs.

Question 30

The duration (in years) of a random pregnancy is a continuous random variable X with the probability density function:

$$f(x) = \begin{cases} ax^6\sqrt{5-6x}, & 0 \leq x \leq \frac{5}{6} \\ 0 & \text{otherwise} \end{cases}$$

- Complete the following sentence: With 90% probability, the **mean** duration of 200 random pregnancies is at least ...months.
- What is the probability that the **mean** duration of 50 random pregnancies deviates by more than half a month from the **median** duration of a single pregnancy?

Tip: Finding the Constant in a Density

For $f(x)$ to be a valid density, the total area under it must equal 1:

$$\int_a^b f(x) dx = 1$$

Enter the formula (without the constant) into Y_1 , use $\text{fnInt}(Y_1, X, a, b)$ to get a number in terms of the constant, then solve for the constant so the total equals 1. We never integrate by hand - always fnInt on the calculator.

Tip: Mean of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **mean** is:

$$\mu = \int_a^b x \cdot f(x) dx = \text{fnInt}(X \cdot Y_1, X, a, b)$$

We never compute this integral by hand - we always use fnInt on the calculator.

💡 Tip: Variance of a Density (via Integration)

For any continuous variable with density $f(x)$ on domain $[a, b]$, the **variance** is:

$$\sigma^2 = \int_a^b x^2 \cdot f(x) dx - \mu^2 = \text{fnInt}(X^2 \cdot Y_1, X, a, b) - \mu^2$$

Just like the mean, we never compute this integral by hand - we always use `fnInt` on the calculator.

💡 Tip: Central Limit Theorem

If we add up or average n independent variables that aren't necessarily normal, the result is only approximately normal, but only if n is large enough:

$$\left. \begin{array}{l} X_1, \dots, X_n \text{ non normal} \\ \text{independent} \\ n \geq 30 \end{array} \right\} \Rightarrow \begin{array}{l} W_n \approx \mathcal{N}(n\mu, \sqrt{n}\sigma) \\ \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \end{array}$$

Notice this is the exact same formula as when the X_i are already normal, just with $\approx$ instead of $\sim$: a single tilde means the result is exact, a double tilde/approx sign means it's only approximated.

If instead we're approximating a Binomial variable with the Normal, an extra condition applies in place of $n \geq 30$: $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$.

✓ **Solution 30: (a) Value below which the mean of 200 pregnancies falls with only 10% probability**

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random pregnancy and record its duration.

- **Mother variable:**

X duration of a random pregnancy (in years)

- **Other random variables:**

$\bar{X}_{200}$ mean duration of 200 random pregnancies (in years)

- **Given:**

$$f(x) = \begin{cases} ax^6\sqrt{5-6x} & 0 \leq x \leq \frac{5}{6} \\ 0 & \text{otherwise} \end{cases}$$

Step 1: Find the constant a .

X is defined on $\left[0, \frac{5}{6}\right]$, so we set the total area under f equal to 1, factor out the constant a , and solve using fnInt:

$$\int_0^{5/6} a \cdot x^6 \sqrt{5-6x} dx = 1 \quad \Rightarrow \quad a \cdot \int_0^{5/6} x^6 \sqrt{5-6x} dx = 1$$

$$a = \frac{1}{\int_0^{5/6} x^6 \sqrt{5-6x} dx} = \frac{1}{\text{fnInt}(x^6 \sqrt{(5-6X)}, X, 0, 5/6)} \approx 35.24$$

Step 2: Find the mean and standard deviation of a single pregnancy's duration.

With a known, we enter the full density into Y_1 , then integrate $x \cdot f(x)$ over the domain to get the mean:

$$Y_1 = 35.24 x^6 \sqrt{5-6x}$$

$$\mu = \text{fnInt}(X \cdot Y_1, X, 0, 5/6) \approx 0.6863$$

Integrate $x^2 \cdot f(x)$ the same way, then subtract μ^2 to get the variance, and take the square root for the standard deviation:

$$E(X^2) = \text{fnInt}(X^2 \cdot Y_1, X, 0, 5/6) \approx 0.4816$$

$$\sigma^2 = 0.4816 - (0.6863)^2 \approx 0.0106$$

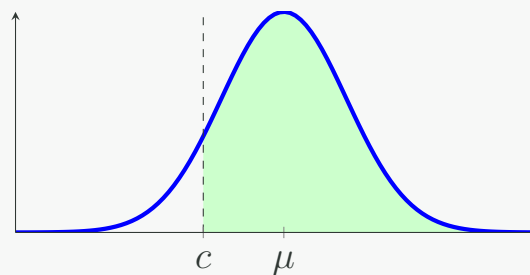
$$\sigma = \sqrt{0.0106} \approx 0.1031$$

Step 3: Find the distribution of the mean duration of 200 pregnancies.

A single pregnancy's duration isn't normal, but we're averaging $n = 200$ of them, and $200 \geq 30$, so the Central Limit Theorem applies:

$$\left. \begin{array}{l} X_1, \dots, X_{200} \text{ non normal} \\ \text{independent} \\ n = 200 \geq 30 \\ E(X) = 0.6863, \sigma_X \approx 0.1031 \end{array} \right\} \Rightarrow \begin{array}{l} \bar{X}_{200} \approx \mathcal{N}\left(0.6863, \frac{0.1031}{\sqrt{200}}\right) \\ = \mathcal{N}(0.6863, 0.007292) \end{array}$$

Step 4: Find the value below which $\bar{X}_{200}$ falls only 10% of the time.



We want the value c such that the shaded area above (90% of the distribution) lies to its right:

$$P(\bar{X}_{200} \geq c) = 0.9$$

We enter this equation directly into the numeric solver, solving for X :

$$\text{normalcdf}(X, \text{E99}, 0.6863, 0.007292) = 0.9$$

$$c \approx 0.6769 \text{ years} \approx 8.123 \text{ months}$$

Answer: With 90% chance, the mean duration of 200 random pregnancies is at least 8.123 months.

Tip: Finding the Median of a Density

The median is the value that splits the total area under $f(x)$ exactly in half - the point where you've swept through 50% of the probability. To find it:

- Set up the equation $\text{fnInt}(Y_1, X, a, M) = 0.5$, where the upper bound M is the unknown we're solving for.
- Enter this equation into the numeric solver and solve for M .

Important: the unknown upper bound must be entered as M (or any letter other than X) - never X itself, since X is already used as the integration variable inside fnInt .

Solution 30: (b) Probability the mean of 50 pregnancies deviates more than half a month from the median

Step 1: Find the median duration of all pregnancies.

We set up the equation $\text{fnInt}(Y_1, X, 0, M) = 0.5$ and solve for M using the numeric solver:

$$\text{Med}(X) \approx 0.7077 \text{ years}$$

Step 2: Translate the event into two one-sided pieces.

Half a month is $\frac{1}{24} \approx 0.0417$ years. "Deviates more than half a month from the median" doesn't say which direction - either $\bar{X}_{50}$ could exceed the median by more than 0.0417, or fall short of it by more than 0.0417. These are two separate, non-overlapping events, so we add their

probabilities instead of using absolute value notation:

$$P(\bar{X}_{50} > 0.7077 + 0.0417) + P(\bar{X}_{50} < 0.7077 - 0.0417)$$

Step 3: Find the distribution of the mean duration of 50 pregnancies.

Using μ and σ from part (a):

$$\bar{X}_{50} \approx \mathcal{N}\left(0.6863, \frac{0.1031}{\sqrt{50}}\right) = \mathcal{N}(0.6863, 0.01458)$$

Step 4: Compute the probability.

Note that $\bar{X}_{50}$ is centered at the mean (0.6863), not at the median (0.7077), so the two tails aren't symmetric here - we compute each one separately:

$$P(\bar{X}_{50} > 0.7494) = \text{normalcdf}(0.7494, E99, 0.6863, 0.01458) \approx 0.0000$$

$$P(\bar{X}_{50} < 0.6660) = \text{normalcdf}(-E99, 0.6660, 0.6863, 0.01458) \approx 0.0825$$

Now we find the total:

$$P(\bar{X}_{50} > 0.7494) + P(\bar{X}_{50} < 0.6660) \approx 0.0000 + 0.0825 = 0.0825$$

Answer: The probability is 0.0825.

? Question 31

Note: The following exercise is not part of the exam material.

It was argued in class that the number of possible random samples of size n , drawn from a finite population of size ν (i.e. there are $\nu \in \mathbb{N}$ members of the population), is

$$\binom{\nu}{n} = \frac{\nu!}{n!(\nu - n)!}$$

if the sampling occurs without replacement and the sampling order is not taken into account. *Random* refers to the fact that all subsets of size n are equally likely to be the sample that is drawn.

It often happens that people combine two samples that were drawn from two different populations. Suppose that a student population of size $\nu = 500$ consists of $\nu_1 = 300$ female students and $\nu_2 = \nu - \nu_1 = 200$ male students. A random sample of size $n_1 = n_2 = 40$ is drawn without replacement from each subpopulation (a subpopulation is called a *stratum*). The combined sample has size $n = n_1 + n_2 = 80$ and is then treated as a random sample without replacement from the entire population of students.

- (a) What is the proportion of samples that are lost by this practice?
- (b) Assume that the variable of interest is the body length of students. What do you expect what happens if the average body length of all students is estimated by the sample average?

Tip: Combinations

The number of ways to choose a sample of size n from a population of size ν without replacement is given by the combination formula:

$$\binom{\nu}{n} = \frac{\nu!}{n!(\nu - n)!}$$

On a TI-84 calculator, this is found using the nCr function (MATH $\rightarrow$ PRB $\rightarrow$ 3:nCr).

Solution 31: (a) What is the proportion of samples that are lost by this practice?

Before computing anything, we set up the framework:

- **Random experiment:** Select a random sample of 80 students from a population of 500, which is divided into two strata.

- **Given:**

$\nu = 500$	total population size
$\nu_1 = 300$	stratum 1 (female) size
$\nu_2 = 200$	stratum 2 (male) size
$n = 80$	total sample size
$n_1 = n_2 = 40$	stratified sample sizes (40 from each stratum)

Step 1: Calculate the total number of possible simple random samples.

If we take a simple random sample of $n = 80$ students from the entire population of $\nu = 500$, the total number of unique samples possible is:

$$N_{SRS} = \binom{500}{80} = 500 \text{ nCr } 80 \approx 1.445 \cdot 10^{94}$$

Step 2: Calculate the total number of possible stratified samples.

With the stratified method, we choose 40 from 300 females AND 40 from 200 males. The total

number of ways to do this is the product of the combinations from each stratum.

$$N_{Stratified} = \binom{300}{40} \cdot \binom{200}{40} = (300 \text{ nCr } 40) \cdot (200 \text{ nCr } 40) \approx 2.008 \cdot 10^{92}$$

Step 3: Calculate the proportion of lost samples.

The stratified method only allows for a small fraction of all possible samples. The proportion of samples that are "lost" (i.e., can never be chosen) by this method is 1 minus the proportion that are possible.

$$\text{Proportion lost} = 1 - \frac{N_{Stratified}}{N_{SRS}} = 1 - \frac{\binom{300}{40} \binom{200}{40}}{\binom{500}{80}} \approx 0.9861$$

This means that by requiring the sample to have exactly 40 females and 40 males, we exclude over 98.6% of all possible groups of 80 students.

Answer: The proportion of samples lost by this practice is approximately 0.9861.

✓ Solution 31: (b) What happens if the average body length of all students is estimated by the sample average?

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random female student and a random male student, and record their body lengths.
- **Mother variables:**

X, Y	body length of a random female student, body length of a random male student
--------	--

• **Other random variables:**

$\bar{X}_{40}$	sample mean body length of the 40 randomly sampled female students
$\bar{Y}_{40}$	sample mean body length of the 40 randomly sampled male students
$\bar{Z}_{80}$	simple average body length of the combined stratified sample of 80 students

When we take a stratified sample with equal numbers from each stratum ($n_1 = 40, n_2 = 40$) and then calculate a simple average, we create a potential for bias if the strata are different sizes.

The simple average of our combined sample, $\bar{Z}_{80}$, would be calculated as:

$$\bar{Z}_{80} = \frac{\text{sum of lengths}}{\text{total students}} = \frac{40\bar{X}_{40} + 40\bar{Y}_{40}}{80} = \frac{\bar{X}_{40} + \bar{Y}_{40}}{2}$$

The expected value of this estimator is:

$$E[\bar{Z}_{80}] = E\left[\frac{\bar{X}_{40} + \bar{Y}_{40}}{2}\right] = \frac{\mu_X + \mu_Y}{2}$$

However, the true population mean, μ_{Total} , is a weighted average based on the actual proportions of females (60%) and males (40%) in the population:

$$\mu_{Total} = \frac{\nu_1\mu_X + \nu_2\mu_Y}{\nu} = \frac{300\mu_X + 200\mu_Y}{500} = 0.6\mu_X + 0.4\mu_Y$$

Because $E[\bar{Z}_{80}] \neq \mu_{Total}$ (unless $\mu_X = \mu_Y$), our sample average is a **biased estimator** of the true population average. It over-weights the male average (50% of the estimate vs. 40% of the population) and under-weights the female average.

Answer: The sample average will be a biased estimator of the population average because the sample proportions (50% female, 50% male) do not match the population proportions (60% female, 40% male).

Question 32

Note: The following exercise is not part of the exam material.

Assume that $(\eta_1, \dots, \eta_\nu)$ are the values of the ν members in the population of a numeric characteristic that we want to investigate. The values in the sample to be drawn are measured by $(X_1, \dots, X_n)$. Notice that we consider the sampling order in this exercise.^a

- In a **random sample with replacement**, the random variables $X_1, \dots, X_n$ are independent and identically distributed, with $P(X_i = \eta_k) = 1/\nu$ for all $i \in \{1, \dots, n\}$ and $k \in \{1, \dots, \nu\}$.
- In a **random sample without replacement**, the random variables $X_1, \dots, X_n$ are dependent but identically distributed, with
 - $P(X_i = \eta_k) = 1/\nu$ for all $i \in \{1, \dots, n\}$ and $k \in \{1, \dots, \nu\}$;
 - $P(X_i = \eta_k, X_j = \eta_l) = 1/(\nu(\nu - 1))$ for all $i, j \in \{1, \dots, n\}$ such that $i \neq j$ and $k, l \in \{1, \dots, \nu\}$ such that $k \neq l$.

The population mean and the population variance are, respectively,

$$\mu = \frac{1}{\nu} \sum_{k=1}^{\nu} \eta_k \quad \text{and} \quad \sigma^2 = \frac{1}{\nu} \sum_{k=1}^{\nu} (\eta_k - \mu)^2.$$

Use the above properties to find $E(X_i)$, $\text{Var}(X_i)$, $\text{Cov}(X_i, X_j)$ for $i \neq j$, $E(\bar{X}_n)$, and $\text{Var}(\bar{X}_n)$ in the case of random sampling:

- (a) with replacement.
- (b) without replacement.

^aDifferent members of the population can have the same value for the considered characteristic, i.e. it is possible that $\eta_k = \eta_l$ when $k \neq l$. This introduces notational problems. Assuming that the members of the population are distinguishable, define $\{X_i = \eta_k\}$ as the event where the i -th observation in the sample is the value that **comes from** the k -th member in the population. For example, if member $k = 15$ and member $l = 204$ have the same value 3, i.e. $\eta_{15} = \eta_{204} = 3$, then the events $\{X_i = \eta_{15}\}$ and $\{X_i = \eta_{204}\}$ are assumed to be disjoint and both to be a subset of $\{X_i = 3\}$. Notice that the image (i.e. set of possible values) of X_i is a (unordered) set containing all different values in $(\eta_1, \dots, \eta_\nu)$.

✓ Solution 32: (a) Sampling with replacement

Before computing anything, we set up the framework:

- **Random experiment:** Draw a random sample of size n from a finite population of size ν and record the value of a numeric characteristic for each sampled member.

- **Mother variable:**

X_i value of the i -th observation in the sample

- **Other random variables:**

$\bar{X}_n$ sample mean of the n observations

- **Given:**

η_k value of the k -th member of the population

$\mu =$ population mean

$\frac{1}{\nu} \sum_{k=1}^{\nu} \eta_k$

$\sigma^2 =$ population variance

$\frac{1}{\nu} \sum_{k=1}^{\nu} (\eta_k - \mu)^2$

.....

In this case, the observations $X_1, \dots, X_n$ are **independent** and identically distributed.

- **$E(X_i)$:** The expected value of any single observation is the population mean.

$$E(X_i) = \sum_{k=1}^{\nu} \eta_k P(X_i = \eta_k) = \sum_{k=1}^{\nu} \eta_k \left(\frac{1}{\nu} \right) = \mu$$

- **$\text{Var}(X_i)$:** The variance of any single observation is the population variance.

$$\text{Var}(X_i) = \sum_{k=1}^{\nu} (\eta_k - \mu)^2 P(X_i = \eta_k) = \sum_{k=1}^{\nu} (\eta_k - \mu)^2 \left(\frac{1}{\nu} \right) = \sigma^2$$

- **Cov(X_i, X_j) for $i \neq j$:** Because the observations are independent, their covariance is zero.

$$\text{Cov}(X_i, X_j) = 0$$

- **$E(\bar{X}_n)$:** The expected value of the sample mean is the population mean.

$$E(\bar{X}_n) = E\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n} \sum E(X_i) = \frac{1}{n}(n\mu) = \mu$$

- **Var($\bar{X}_n$):** The variance of the sample mean is the population variance divided by n .

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n^2} \sum \text{Var}(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}$$

Answer: $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2$, $\text{Cov}(X_i, X_j) = 0$, $E(\bar{X}_n) = \mu$, and $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$.

✓ Solution 32: (b) Sampling without replacement

In this case, the observations $X_1, \dots, X_n$ are **dependent**.

- **$E(X_i)$:** The derivation is identical to the case with replacement. Each draw is still from the same overall population.

$$E(X_i) = \mu$$

- **Var(X_i):** The derivation is also identical to the case with replacement.

$$\text{Var}(X_i) = \sigma^2$$

- **Cov(X_i, X_j) for $i \neq j$:** Because the observations are dependent, their covariance is non-zero. It can be shown to be:

$$\text{Cov}(X_i, X_j) = -\frac{\sigma^2}{n-1}$$

- **$E(\bar{X}_n)$:** The derivation is identical to the case with replacement, as the linearity of expect-

tation does not require independence.

$$E(\bar{X}_n) = \mu$$

- **Var($\bar{X}_n$):** The variance of the sample mean now includes the covariance terms.

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n^2} \left(\sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \right) \\ &= \frac{1}{n^2} \left(n\sigma^2 + n(n-1) \left(-\frac{\sigma^2}{\nu-1} \right) \right) \\ &= \frac{\sigma^2}{n} \left(1 - \frac{n-1}{\nu-1} \right) = \frac{\sigma^2}{n} \left(\frac{\nu-n}{\nu-1} \right) \end{aligned}$$

The term $\left(\frac{\nu-n}{\nu-1} \right)$ is known as the **finite-population correction factor**.

Answer: $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2$, $\text{Cov}(X_i, X_j) = -\frac{\sigma^2}{\nu-1}$, $E(\bar{X}_n) = \mu$, and $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \left(\frac{\nu-n}{\nu-1} \right)$.

Chapter 2 - Confidence Intervals

Chapter 3 - Testing One Population Parameter

Chapter 4 - Testing Two Population Parameters

Chapter 5 - One-Way ANOVA

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